Three-model single-graph edge and bulk evidence synthesis

Non-Perron localization, spacing, support geometry, and random-set controls

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Evidence status: comprehensive but fixed-realization. One stored $N = 5000$, $m = 10$ graph is analyzed for each model. The report adds top-15 and top-20 summaries, uses consistent non-Perron edge windows, matched centered zero-eigenvalue-free bulk windows, and the existing 1,000-set controls. It does not turn three graphs into ensemble inference.

1 Executive conclusions

For each nested upper-edge window $K = 10, 15, 20, 25, 50$, we examine every ranked non-Perron eigenvalue, its corresponding eigenvector, and the ten vertices carrying that eigenvector's largest squared-coordinate masses. The edge eigenvectors are more concentrated than their bulk counterparts according to IPR, N IPR, and M_{10}, the mass contained in the ten largest coordinates. Their top-mass vertices preferentially select hubs, form denser induced subgraphs, and lie at shorter full-graph distances than bulk or uniformly random ten-vertex supports. By contrast, the gap-ratio comparison provides no strong, uniform evidence that edge and bulk levels behave substantially differently: for nonlinear $\zeta = 1.5$, the narrowest edge window is near Poisson, but broader windows move toward GOE-like values. Eigenvector localization is therefore the more consistent edge-bulk signal.

Scope. Nested windows are descriptive and reuse levels. There are no graph-level confidence intervals, parameter sweep, finite-size scaling, diagonal disorder, or random Schrödinger operator.

2 Definitions and window construction

For unit v, IPR $= \sum_j |v_j|^4$, N IPR $= N \sum_j |v_j|^4$, and M_{10} is the sum of its ten largest masses. For an interior level, $r = \min(s_L, s_R)/\max(s_L, s_R)$. Edge-$K$ contains non-Perron ranks $2, \ldots, K+1$; strict r uses ranks $3, \ldots, K$, hence $K-2$ ratios. Bulk-K is centered on the recorded zero-eigenvalue-free center and also excludes its two endpoints only for r. Every eigenvector/support summary uses all K states.

Exact-zero eigenvalues (tolerance 10^{-8}) remain in density figures but are removed from continuous bulk and gap statistics because repeated zero eigenvalues yield zero gaps, undefined 0/0 ratios, and basis-dependent individual eigenvectors within the zero-eigenvalue subspace.

3 Provenance and disclosed extension

The fixed graphs, dense eigenvalues, full-spectrum IPR, selected eigenvectors, baselines, and original plots were reused. The original edge vectors ended at rank 50. To complete non-Perron top 50 (ranks 2–51), rank 51 was recovered with a 55-vector sparse largest-algebraic solve. Top-50 eigenvalues matched the dense spectrum within 6.3×10^{-13} and rank-51 residuals were at most 2.5×10^{-14}. See `tables/rank51_sparse_validation.csv`.

4 Cross-model headline comparison

Model	K	Edge $\bar{r}$	Bulk $\bar{r}$	Edge NIPR	Bulk NIPR	Edge M_{10}	Bulk M_{10}	Edge dens.	Bulk dens.
affine $\alpha = 1.05$	10	0.6321	0.5398	267.67	42.25	0.4700	0.2176	0.8867	0.0244
affine $\alpha = 1.05$	15	0.6429	0.6015	342.51	40.54	0.4708	0.2132	0.8859	0.0296
affine $\alpha = 1.05$	20	0.5889	0.6001	498.15	37.92	0.4733	0.2056	0.8044	0.0278
affine $\alpha = 1.05$	25	0.5865	0.6014	570.06	37.66	0.4767	0.2060	0.6880	0.0267
affine $\alpha = 1.05$	50	0.5434	0.5409	499.24	49.89	0.4690	0.2310	0.4156	0.0258
nonlinear $\zeta = 1.5$	10	0.3868	0.4908	389.30	10.53	0.4866	0.0804	0.8733	0.0000
nonlinear $\zeta = 1.5$	15	0.4264	0.5239	421.68	10.61	0.4837	0.0801	0.7585	0.0000
nonlinear $\zeta = 1.5$	20	0.4676	0.5578	397.19	10.54	0.4824	0.0787	0.6456	0.0000
nonlinear $\zeta = 1.5$	25	0.4909	0.5955	373.18	10.62	0.4770	0.0788	0.5796	0.0000
nonlinear $\zeta = 1.5$	50	0.5481	0.5907	311.68	10.75	0.4345	0.0798	0.3733	0.0000
affine $\alpha = 2.0$	10	0.4760	0.4345	155.30	3.86	0.3855	0.0333	0.5867	0.0000
affine $\alpha = 2.0$	15	0.5644	0.4563	136.58	3.84	0.3684	0.0328	0.6015	0.0000
affine $\alpha = 2.0$	20	0.5281	0.4520	119.78	3.84	0.3507	0.0331	0.5544	0.0000
affine $\alpha = 2.0$	25	0.5125	0.4488	106.29	3.84	0.3352	0.0331	0.5013	0.0000
affine $\alpha = 2.0$	50	0.5167	0.4357	72.79	3.84	0.2737	0.0331	0.3347	0.0000

Table 1: Strict mean gap ratios and mean localization/support density.

Column guide. K is the number of eigenvectors in the window; each eigenvector still contributes a separate support of exactly ten top-mass vertices. Edge $\bar{r}$ and Bulk $\bar{r}$ are the means of the eligible raw adjacent-gap ratios $r_i = \min(s_{i-1}, s_i)/\max(s_{i-1}, s_i)$ in their respective windows. Edge/Bulk NIPR are window means of $N\sum_j |v_j|^4$, so larger values indicate stronger eigenvector localization. Edge/Bulk M_{10} are window means of the mass $\sum_{j\in S_i} |v_j|^2$ on each eigenvector's ten largest-mass coordinates. Edge/Bulk dens. are window means of $|E(G[S_i])|/\binom{10}{2} = |E(G[S_i])|/45$.

5 Detailed window summaries: affine $\alpha = 1.05$

5.1 Spectral and eigenvector statistics

Column guide. Region identifies the upper spectral edge or centered bulk; K is the number of eigenvectors summarized; and n_r is the number of eligible interior gap ratios. $\bar{r}$/med. gives the mean/median of those n_r raw ratios. IPR mean/med. summarize $\sum_j |v_j|^4$ across the K eigenvectors, and NIPR mean/med. summarize the same quantity multiplied by $N = 5000$. M_{10} mean/med. summarize, across the K eigenvectors, the total mass on each eigenvector's ten largest-mass coordinates. Degree mean/med. are the mean/median across the K eigenvectors of the average full-graph degree of those ten selected vertices. Thus every mean/median pair except $\bar{r}$/med. is taken across K eigenvector-level measurements.

For the bulk rows, exact-zero eigenvalues are first removed at tolerance 10^{-8} so that neither zero gaps nor undefined 0/0 ratios enter the selection. The remaining eigenvalues are split into contiguous negative- and positive-energy segments. The bulk-50 center is the feasible center closest to the median rank of this zero-eigenvalue-free spectrum, with one neighboring level available on each side; bulk-10, 15, 20, and 25 are nested around that same center. Only the two endpoints of each displayed bulk-K window are omitted from its strict gap-ratio summary.

Region	K	n_r	$\bar{r}$/med.	IPR mean	IPR med.	NIPR mean	NIPR med.	M_{10} mean	M_{10} med.	degree mean	degree med.
edge	10	8	0.6321/0.6862	0.053534	0.038504	267.67	192.52	0.4700	0.4701	1873.11	1894.95
edge	15	13	0.6429/0.5970	0.068502	0.047170	342.51	235.85	0.4708	0.4698	1836.98	1857.00
edge	20	18	0.5889/0.5726	0.099631	0.065987	498.15	329.93	0.4733	0.4749	1696.55	1805.90
edge	25	23	0.5865/0.5819	0.114013	0.144604	570.06	723.02	0.4767	0.4800	1402.97	1768.80
edge	50	48	0.5434/0.5779	0.099848	0.076281	499.24	381.41	0.4690	0.4719	764.52	171.85
bulk	10	8	0.5398/0.5943	0.008450	0.007286	42.25	36.43	0.2176	0.2042	10.50	10.40
bulk	15	13	0.6015/0.6641	0.008107	0.007555	40.54	37.78	0.2132	0.2030	10.50	10.50
bulk	20	18	0.6001/0.6319	0.007584	0.007286	37.92	36.43	0.2056	0.1997	10.52	10.50
bulk	25	23	0.6014/0.5996	0.007531	0.007555	37.66	37.78	0.2060	0.2030	10.53	10.50
bulk	50	48	0.5409/0.5591	0.009978	0.007699	49.89	38.50	0.2310	0.2042	10.57	10.55

5.2 Induced subgraph and full-graph connectivity

For each eigenvector, S_i contains its ten largest-mass vertices and $G[S_i]$ is their induced subgraph. Edges mean/med. summarize $|E(G[S_i])|$ across the K supports. Density mean/med. summarize $|E(G[S_i])|/45$. Comp. mean/med. summarize the number of connected components in $G[S_i]$, and LCC mean/med. summarize the number of vertices in its largest connected component (between 1 and 10, regardless of K). For each support, distance is the mean of the 45 pairwise shortest-path distances measured in the full graph, allowing paths through vertices outside S_i; distance mean/med. then summarize those K support-level means. Frac. ≤ 2 and frac. ≤ 3 are the means across the K supports of the fractions of their 45 vertex pairs whose full-graph distances are at most two or three. The repeated *med.* heading always refers to the immediately preceding metric.

Region	K	edges mean	med.	density mean	med.	comp. mean	med.	LCC mean	med.	distance mean	med.	frac. ≤ 2	frac. ≤ 3
edge	10	39.90	40.00	0.8867	0.8889	1.00	1.00	10.00	10.00	1.1133	1.1111	1.0000	1.0000
edge	15	39.87	40.00	0.8859	0.8889	1.00	1.00	10.00	10.00	1.1141	1.1111	1.0000	1.0000
edge	20	36.20	39.50	0.8044	0.8778	1.00	1.00	10.00	10.00	1.1956	1.1222	1.0000	1.0000
edge	25	30.96	39.00	0.6880	0.8667	1.40	1.00	9.60	10.00	1.3120	1.1333	1.0000	1.0000
edge	50	18.70	11.50	0.4156	0.2556	3.08	2.00	7.72	9.00	1.5862	1.7444	0.9982	1.0000
bulk	10	1.10	1.00	0.0244	0.0222	8.90	9.00	2.00	2.00	2.0467	2.0333	0.9289	1.0000
bulk	15	1.33	1.00	0.0296	0.0222	8.67	9.00	2.20	2.00	2.0430	2.0444	0.9274	1.0000
bulk	20	1.25	1.00	0.0278	0.0222	8.75	9.00	2.10	2.00	2.0489	2.0444	0.9233	1.0000
bulk	25	1.20	1.00	0.0267	0.0222	8.80	9.00	2.08	2.00	2.0444	2.0444	0.9289	1.0000
bulk	50	1.16	1.00	0.0258	0.0222	8.84	9.00	2.08	2.00	2.0538	2.0444	0.9204	1.0000

6 Detailed window summaries: nonlinear $\zeta = 1.5$

6.1 Spectral and eigenvector statistics

Region	K	n_r	$\bar{r}$/med.	IPR mean	IPR med.	NIPR mean	NIPR med.	M_{10} mean	M_{10} med.	degree mean	degree med.
edge	10	8	0.3868/0.3040	0.077860	0.071707	389.30	358.53	0.4866	0.4869	2195.10	2237.20
edge	15	13	0.4264/0.3939	0.084335	0.082465	421.68	412.32	0.4837	0.4866	1733.13	2029.10
edge	20	18	0.4676/0.4090	0.079437	0.072741	397.19	363.71	0.4824	0.4864	1394.97	1801.70
edge	25	23	0.4909/0.4723	0.074636	0.064419	373.18	322.10	0.4770	0.4856	1179.12	496.10
edge	50	48	0.5481/0.5307	0.062336	0.051570	311.68	257.85	0.4345	0.4496	646.22	258.90
bulk	10	8	0.4908/0.3682	0.002106	0.002105	10.53	10.52	0.0804	0.0798	10.07	10.10
bulk	15	13	0.5239/0.5658	0.002122	0.002156	10.61	10.78	0.0801	0.0795	10.07	10.10
bulk	20	18	0.5578/0.5896	0.002109	0.002026	10.54	10.13	0.0787	0.0775	10.06	10.10
bulk	25	23	0.5955/0.6208	0.002124	0.002054	10.62	10.27	0.0788	0.0785	10.07	10.10
bulk	50	48	0.5907/0.6166	0.002151	0.002061	10.75	10.31	0.0798	0.0797	10.08	10.00

6.2 Induced subgraph and full-graph connectivity

Region	K	edges mean	med.	density mean	med.	comp. mean	med.	LCC mean	med.	distance mean	med.	frac. ≤ 2	frac. ≤ 3
edge	10	39.30	40.50	0.8733	0.9000	1.00	1.00	10.00	10.00	1.1267	1.1000	1.0000	1.0000
edge	15	34.13	38.00	0.7585	0.8444	1.00	1.00	10.00	10.00	1.2415	1.1556	1.0000	1.0000
edge	20	29.05	34.00	0.6456	0.7556	1.15	1.00	9.80	10.00	1.3544	1.2444	1.0000	1.0000
edge	25	26.08	22.00	0.5796	0.4889	1.12	1.00	9.84	10.00	1.4204	1.5111	1.0000	1.0000
edge	50	16.80	12.50	0.3733	0.2778	2.50	2.00	7.86	9.00	1.6307	1.7222	0.9960	1.0000
bulk	10	0.00	0.00	0.0000	0.0000	10.00	10.00	1.00	1.00	2.0000	2.0000	1.0000	1.0000
bulk	15	0.00	0.00	0.0000	0.0000	10.00	10.00	1.00	1.00	2.0000	2.0000	1.0000	1.0000
bulk	20	0.00	0.00	0.0000	0.0000	10.00	10.00	1.00	1.00	2.0000	2.0000	1.0000	1.0000
bulk	25	0.00	0.00	0.0000	0.0000	10.00	10.00	1.00	1.00	2.0000	2.0000	1.0000	1.0000
bulk	50	0.00	0.00	0.0000	0.0000	10.00	10.00	1.00	1.00	2.0036	2.0000	0.9964	1.0000

7 Detailed window summaries: affine $\alpha = 2.0$

7.1 Spectral and eigenvector statistics

Region	K	n_r	$\bar{r}$/med.	IPR mean	IPR med.	NIPR mean	NIPR med.	M_{10} mean	M_{10} med.	degree mean	degree med.
edge	10	8	0.4760/0.3341	0.031060	0.029470	155.30	147.35	0.3855	0.3918	245.30	245.20
edge	15	13	0.5644/0.5377	0.027316	0.024258	136.58	121.29	0.3684	0.3511	238.94	239.10
edge	20	18	0.5281/0.4719	0.023957	0.018913	119.78	94.57	0.3507	0.3453	229.11	230.40
edge	25	23	0.5125/0.4361	0.021259	0.016687	106.29	83.44	0.3352	0.3302	220.71	225.80
edge	50	48	0.5167/0.4882	0.014558	0.009980	72.79	49.90	0.2737	0.2637	175.00	174.60
bulk	10	8	0.4345/0.4462	0.000772	0.000756	3.86	3.78	0.0333	0.0312	10.61	10.60
bulk	15	13	0.4563/0.5552	0.000768	0.000761	3.84	3.81	0.0328	0.0314	10.63	10.60
bulk	20	18	0.4520/0.4255	0.000768	0.000762	3.84	3.81	0.0331	0.0317	10.75	10.65
bulk	25	23	0.4488/0.3705	0.000767	0.000763	3.84	3.82	0.0331	0.0320	10.74	10.60
bulk	50	48	0.4357/0.3986	0.000768	0.000765	3.84	3.83	0.0331	0.0330	10.77	10.70

7.2 Induced subgraph and full-graph connectivity

Region	K	edges mean	med.	density mean	med.	comp. mean	med.	LCC mean	med.	distance mean	med.	frac. ≤ 2	frac. ≤ 3
edge	10	26.40	26.50	0.5867	0.5889	1.00	1.00	10.00	10.00	1.4133	1.4111	1.0000	1.0000
edge	15	27.07	27.00	0.6015	0.6000	1.00	1.00	10.00	10.00	1.3985	1.4000	1.0000	1.0000
edge	20	24.95	25.00	0.5544	0.5556	1.15	1.00	9.85	10.00	1.4456	1.4444	1.0000	1.0000
edge	25	22.56	25.00	0.5013	0.5556	1.28	1.00	9.68	10.00	1.4987	1.4444	1.0000	1.0000
edge	50	15.06	11.00	0.3347	0.2444	2.42	2.00	8.04	9.00	1.6684	1.7556	0.9969	1.0000
bulk	10	0.00	0.00	0.0000	0.0000	10.00	10.00	1.00	1.00	2.9289	2.9111	0.1000	0.9711
bulk	15	0.00	0.00	0.0000	0.0000	10.00	10.00	1.00	1.00	2.9481	2.9333	0.0933	0.9585
bulk	20	0.00	0.00	0.0000	0.0000	10.00	10.00	1.00	1.00	2.9444	2.9444	0.0978	0.9578
bulk	25	0.00	0.00	0.0000	0.0000	10.00	10.00	1.00	1.00	2.9396	2.9333	0.0996	0.9609
bulk	50	0.00	0.00	0.0000	0.0000	10.00	10.00	1.00	1.00	2.9316	2.9111	0.1071	0.9613

9 Numerical interpretation and random-support controls

The spacing-ratio conclusions are weaker than the eigenvector conclusions. Edge and bulk strict mean r overlap in several windows, so these three fixed graphs do not support a universal edge-Poisson claim. In contrast, every window size shows substantially larger edge IPR, N IPR, and M_{10}, with hub-heavy, dense, compact top-mass supports.

Induced density is the number of displayed edges divided by $\binom{10}{2} = 45$. Components and LCC are properties of the ten-vertex induced graph; LCC is the number of vertices in its largest connected component. Distances are shortest paths in the full graph. For each model, 1,000 uniformly sampled ten-vertex sets define $z = (T_{\text{obs}} - \bar{T}_{\text{rand}})/s_{\text{rand}}$ and the inclusive percentile. Negative distance z means compact; positive density or degree z means unusually connected or hub-heavy. These controls are not degree-matched, and z-scores are not Gaussian p-values.

10 Random 1,000-set reference distributions

For each model's same fixed graph, a seeded random generator draws 1,000 sets independently. Each draw samples ten distinct vertices uniformly without replacement within that set; different draws may overlap. The same degree, induced-subgraph, connectivity, and full-graph distance statistics used for an eigenvector's top-ten support are computed for every random set. These controls are uniform rather than degree-matched and contain no eigenvector-based selection.

Model	Metric	n	Mean	Median	SD	Min	Max
affine $\alpha = 1.05$	max_degree	1000	95.1530	19.0000	319.8786	10.0000	2168.0000
affine $\alpha = 1.05$	mean_degree	1000	19.6009	11.5000	34.2652	10.0000	384.5000
affine $\alpha = 1.05$	median_degree	1000	10.2205	10.0000	0.4537	10.0000	14.0000
affine $\alpha = 1.05$	induced_edge_count	1000	0.1570	0.0000	0.6744	0.0000	7.0000
affine $\alpha = 1.05$	induced_density	1000	0.0035	0.0000	0.0150	0.0000	0.1556
affine $\alpha = 1.05$	induced_num_components	1000	9.8430	10.0000	0.6744	3.0000	10.0000
affine $\alpha = 1.05$	induced_largest_component	1000	1.1540	1.0000	0.6608	1.0000	8.0000
affine $\alpha = 1.05$	mean_pairwise_distance	1000	2.0479	2.0444	0.0450	1.8667	2.2444
affine $\alpha = 1.05$	median_pairwise_distance	1000	2.0000	2.0000	0.0000	2.0000	2.0000
affine $\alpha = 1.05$	fraction_distance_1	1000	0.0035	0.0000	0.0150	0.0000	0.1556
affine $\alpha = 1.05$	fraction_distance_le_2	1000	0.9486	0.9556	0.0425	0.7556	1.0000
affine $\alpha = 1.05$	fraction_distance_le_3	1000	1.0000	1.0000	0.0000	1.0000	1.0000
nonlinear $\zeta = 1.5$	max_degree	1000	82.3780	19.0000	345.9587	11.0000	3672.0000
nonlinear $\zeta = 1.5$	mean_degree	1000	18.9680	12.3000	37.8790	10.1000	520.8000

Model	Metric	n	Mean	Median	SD	Min	Max
nonlinear $\zeta = 1.5$	median_degree	1000	11.0905	11.0000	0.8870	10.0000	18.5000
nonlinear $\zeta = 1.5$	induced_edge_count	1000	0.1550	0.0000	0.7277	0.0000	9.0000
nonlinear $\zeta = 1.5$	induced_density	1000	0.0034	0.0000	0.0162	0.0000	0.2000
nonlinear $\zeta = 1.5$	induced_num_components	1000	9.8460	10.0000	0.7175	1.0000	10.0000
nonlinear $\zeta = 1.5$	induced_largest_component	1000	1.1540	1.0000	0.7175	1.0000	10.0000
nonlinear $\zeta = 1.5$	mean_pairwise_distance	1000	2.0194	2.0000	0.0378	1.8000	2.3333
nonlinear $\zeta = 1.5$	median_pairwise_distance	1000	2.0000	2.0000	0.0000	2.0000	2.0000
nonlinear $\zeta = 1.5$	fraction_distance_1	1000	0.0034	0.0000	0.0162	0.0000	0.2000
nonlinear $\zeta = 1.5$	fraction_distance_le_2	1000	0.9772	1.0000	0.0336	0.6444	1.0000
nonlinear $\zeta = 1.5$	fraction_distance_le_3	1000	1.0000	1.0000	0.0000	1.0000	1.0000
affine $\alpha = 2.0$	max_degree	1000	55.8480	38.0000	49.7617	14.0000	355.0000
affine $\alpha = 2.0$	mean_degree	1000	19.6560	17.9000	6.7085	10.9000	66.2000
affine $\alpha = 2.0$	median_degree	1000	14.3855	14.0000	2.4233	10.0000	31.0000
affine $\alpha = 2.0$	induced_edge_count	1000	0.1660	0.0000	0.4250	0.0000	4.0000
affine $\alpha = 2.0$	induced_density	1000	0.0037	0.0000	0.0094	0.0000	0.0889
affine $\alpha = 2.0$	induced_num_components	1000	9.8350	10.0000	0.4171	7.0000	10.0000
affine $\alpha = 2.0$	induced_largest_component	1000	1.1560	1.0000	0.3871	1.0000	4.0000
affine $\alpha = 2.0$	mean_pairwise_distance	1000	2.9123	2.9111	0.1062	2.4667	3.2222
affine $\alpha = 2.0$	median_pairwise_distance	1000	2.9990	3.0000	0.0316	2.0000	3.0000
affine $\alpha = 2.0$	fraction_distance_1	1000	0.0037	0.0000	0.0094	0.0000	0.0889
affine $\alpha = 2.0$	fraction_distance_le_2	1000	0.1392	0.1333	0.0771	0.0000	0.5111
affine $\alpha = 2.0$	fraction_distance_le_3	1000	0.9448	0.9556	0.0476	0.7333	1.0000

11 Window comparisons with random supports

Entries are mean z-score / mean inclusive percentile.

Model	Region	K	Distance	Induced density	Maximum degree	Mean degree
affine $\alpha = 1.05$	edge	10	-20.77/0.00	58.93/100.00	6.33/99.92	54.09/100.00
affine $\alpha = 1.05$	edge	15	-20.75/0.00	58.88/100.00	6.36/99.93	53.04/100.00
affine $\alpha = 1.05$	edge	20	-18.94/0.00	53.45/100.00	6.30/99.86	48.94/100.00
affine $\alpha = 1.05$	edge	25	-16.35/0.00	45.68/100.00	5.30/99.14	40.37/99.85
affine $\alpha = 1.05$	edge	50	-10.26/0.24	27.50/99.66	2.83/96.63	21.74/98.45
affine $\alpha = 1.05$	bulk	10	-0.03/55.29	1.40/95.23	-0.26/17.39	-0.27/18.24
affine $\alpha = 1.05$	bulk	15	-0.11/55.35	1.74/95.76	-0.26/17.77	-0.27/18.22
affine $\alpha = 1.05$	bulk	20	0.02/58.29	1.62/95.80	-0.26/18.32	-0.27/19.09
affine $\alpha = 1.05$	bulk	25	-0.08/55.62	1.55/95.67	-0.26/18.03	-0.26/19.53
affine $\alpha = 1.05$	bulk	50	0.13/59.16	1.49/95.73	-0.26/18.24	-0.26/20.76
nonlinear $\zeta = 1.5$	edge	10	-23.64/0.00	53.79/100.00	10.15/99.93	57.45/100.00
nonlinear $\zeta = 1.5$	edge	15	-20.60/0.00	46.69/100.00	8.60/99.57	45.25/99.95
nonlinear $\zeta = 1.5$	edge	20	-17.61/0.00	39.71/100.00	6.78/99.07	36.33/99.89
nonlinear $\zeta = 1.5$	edge	25	-15.86/0.00	35.63/100.00	5.64/98.67	30.63/99.79
nonlinear $\zeta = 1.5$	edge	50	-10.30/0.29	22.87/99.84	2.95/96.50	16.56/98.87
nonlinear $\zeta = 1.5$	bulk	10	-0.51/52.70	-0.21/91.80	-0.21/0.42	-0.23/0.14
nonlinear $\zeta = 1.5$	bulk	15	-0.51/52.70	-0.21/91.80	-0.21/0.40	-0.23/0.15
nonlinear $\zeta = 1.5$	bulk	20	-0.51/52.70	-0.21/91.80	-0.21/0.36	-0.24/0.13
nonlinear $\zeta = 1.5$	bulk	25	-0.51/52.70	-0.21/91.80	-0.21/0.34	-0.23/0.17
nonlinear $\zeta = 1.5$	bulk	50	-0.42/54.46	-0.21/91.80	-0.21/0.66	-0.23/0.27
affine $\alpha = 2.0$	edge	10	-14.12/0.00	61.73/100.00	5.26/99.81	33.64/100.00
affine $\alpha = 2.0$	edge	15	-14.26/0.00	63.30/100.00	5.09/99.73	32.69/100.00
affine $\alpha = 2.0$	edge	20	-13.82/0.00	58.32/100.00	4.92/99.63	31.22/100.00
affine $\alpha = 2.0$	edge	25	-13.32/0.00	52.69/100.00	4.71/99.39	29.97/100.00
affine $\alpha = 2.0$	edge	50	-11.72/0.00	35.04/100.00	3.58/97.57	23.16/100.00
affine $\alpha = 2.0$	bulk	10	0.16/57.26	-0.39/85.20	-0.87/0.19	-1.35/0.03
affine $\alpha = 2.0$	bulk	15	0.34/62.36	-0.39/85.20	-0.87/0.13	-1.34/0.03
affine $\alpha = 2.0$	bulk	20	0.30/61.84	-0.39/85.20	-0.86/0.29	-1.33/0.06
affine $\alpha = 2.0$	bulk	25	0.26/60.79	-0.39/85.20	-0.86/0.26	-1.33/0.06
affine $\alpha = 2.0$	bulk	50	0.18/58.85	-0.39/85.20	-0.86/0.34	-1.33/0.06

12 Interpretation of the original diagnostic figures

The interpretations below refer to the original recorded plots, not to newly drawn window-summary curves. No new diagnostic figure is introduced in this revision. The three eigenvalue-density panels receive only a terminology-only

title relabel; their underlying data are unchanged. The figures display edge ranks 1–50 including the Perron state, whereas the numerical edge tables use non-Perron ranks 2 onward. This distinction is kept explicit.

12.1 How to read the four-panel eigenpair microscopes

The upper-left panel reports an *individual* adjacent-gap ratio for each eligible eigenvalue, not a moving mean. The dashed GOE and dotted Poisson lines are ensemble reference means, so a single point crossing either line is not a classification. The upper-right panel reports N IPR on a logarithmic scale; higher means more localization. The lower-left panel reports M_{10}, the sum of the ten largest squared eigenvector coordinates; it is not a distance. The lower-right panel asks whether spacing ratio and localization co-vary. Across all three graphs they do not do so monotonically.

12.2 How to read the spacing histograms

Raw gaps depend on the local spectral density, so raw histograms from different energy bands cannot be compared as though they share one physical scale. The unfolded plots rescale the local mean spacing to one and are the appropriate visual comparison with GOE and Poisson reference curves. Extreme-tail panels contain the fewest gaps and therefore have the largest finite-sample noise. These are one-graph diagnostics, not ensemble tests.

12.3 How to read the induced-support drawings

Each drawing keeps only the ten vertices carrying the largest eigenvector masses. Node area encodes mass, color encodes full-graph degree, and black lines are only edges among those ten vertices. Missing black lines do not mean that two vertices are far apart in the full graph; full-graph shortest-path distances can pass through omitted vertices and are summarized separately. Edge supports form visible hub-rich cores, while the centered bulk supports are mostly isolated in the induced drawings.

13 Recorded spacing and edge-bulk diagnostics

13.1 affine $\alpha = 1.05$

Eigenvalue density. The left panel makes the large block of repeated zero eigenvalues visible, while the right panel removes the zero-eigenvalue modes and the Perron root to reveal the continuous levels used for gap statistics. The narrow central spike explains why raw gaps near zero are small; it is a density effect, not by itself evidence that the corresponding eigenvectors are extended or localized.

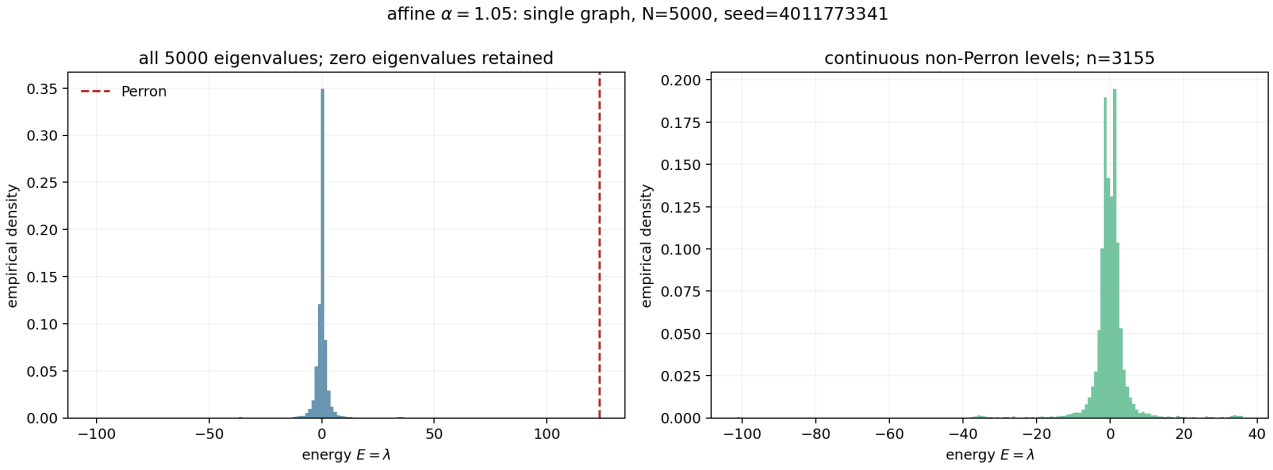


Figure 1: Original single-graph eigenvalue-density diagnostic for affine $\alpha = 1.05$.

Raw and unfolded local spacing. The raw histograms vary strongly across energy because the spectral density varies. After local unfolding, most populated interior bins are qualitatively closer to the GOE curve than to the Poisson curve, while the extreme bins contain few gaps and are visibly noisy. The global ratio mean is about 0.531, close to GOE, but that full-spectrum average does not resolve the rank-specific edge fluctuations shown by the microscope.

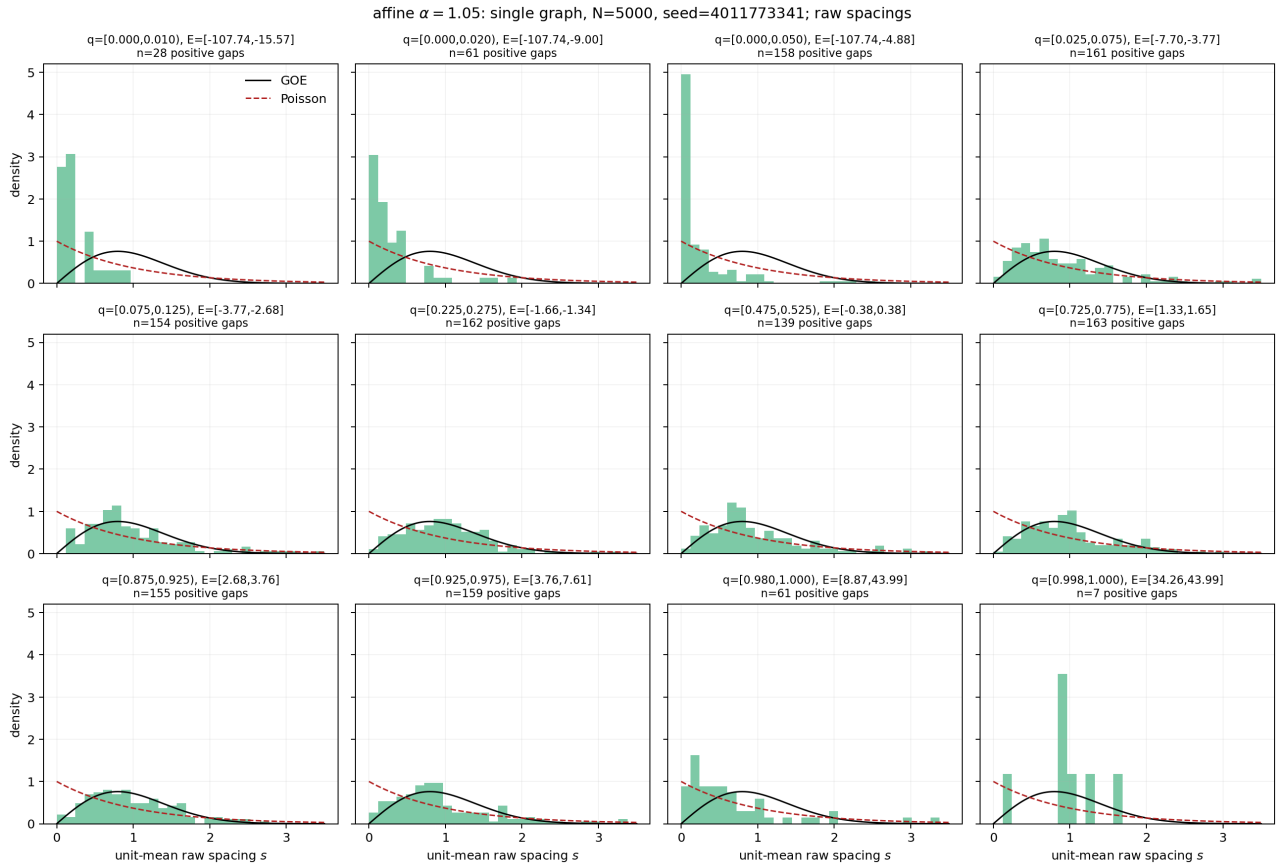


Figure 2: Original local raw-spacing histograms for affine $\alpha = 1.05$.

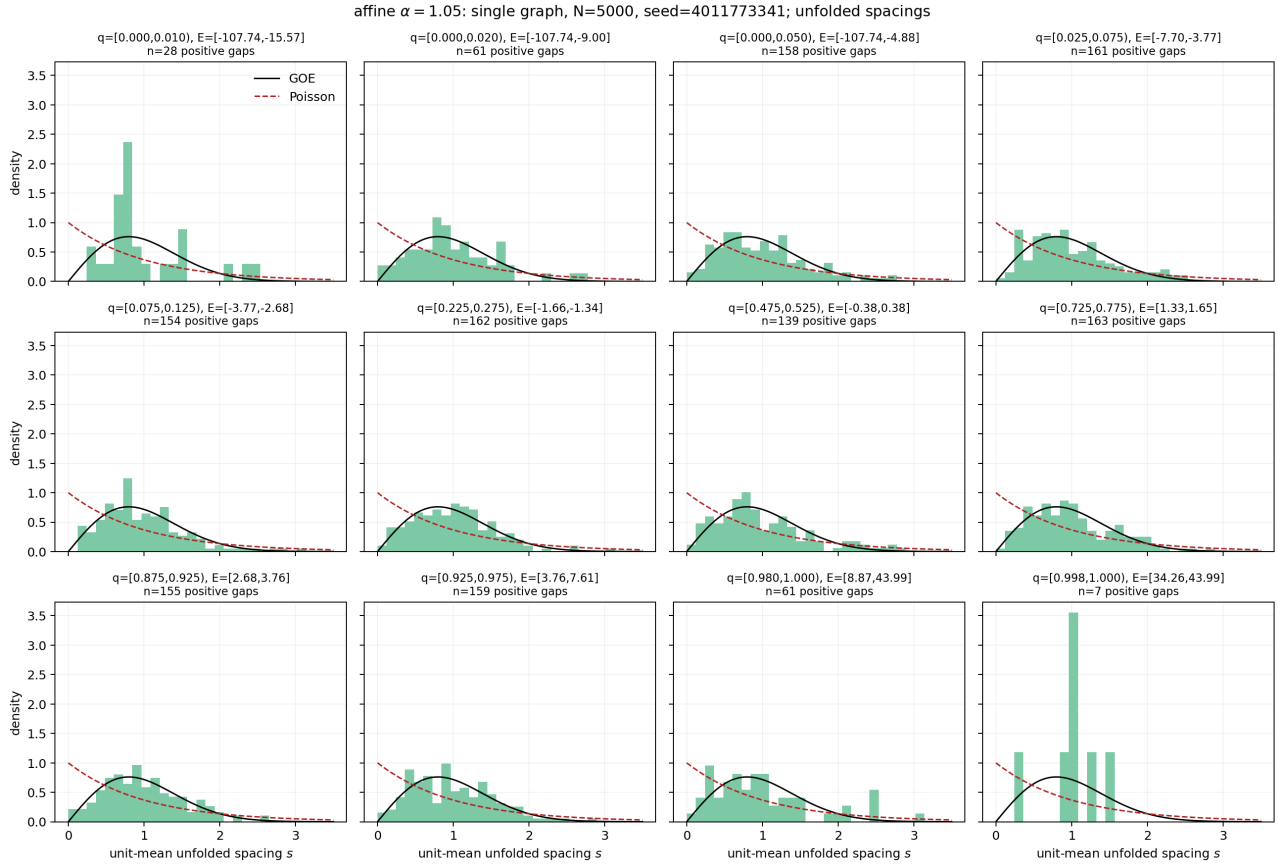


Figure 3: Original local unfolded-spacing histograms for affine $\alpha = 1.05$.

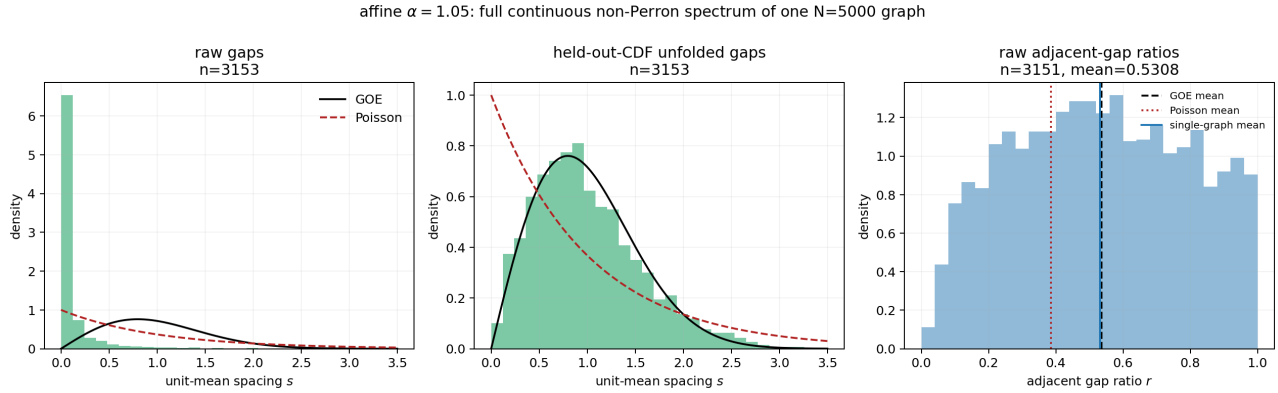


Figure 4: Original full continuous-spectrum spacing and ratio diagnostic for affine $\alpha = 1.05$.

Localization versus energy. The energy–N IPR cloud has a pronounced localization bowl: the continuous bulk near $E = 0$ is least localized, whereas the spectral tails rise by one to two orders of magnitude. The positive-edge rings sit high in the cloud; the isolated far-right Perron point must not be mixed with the non-Perron window summaries.

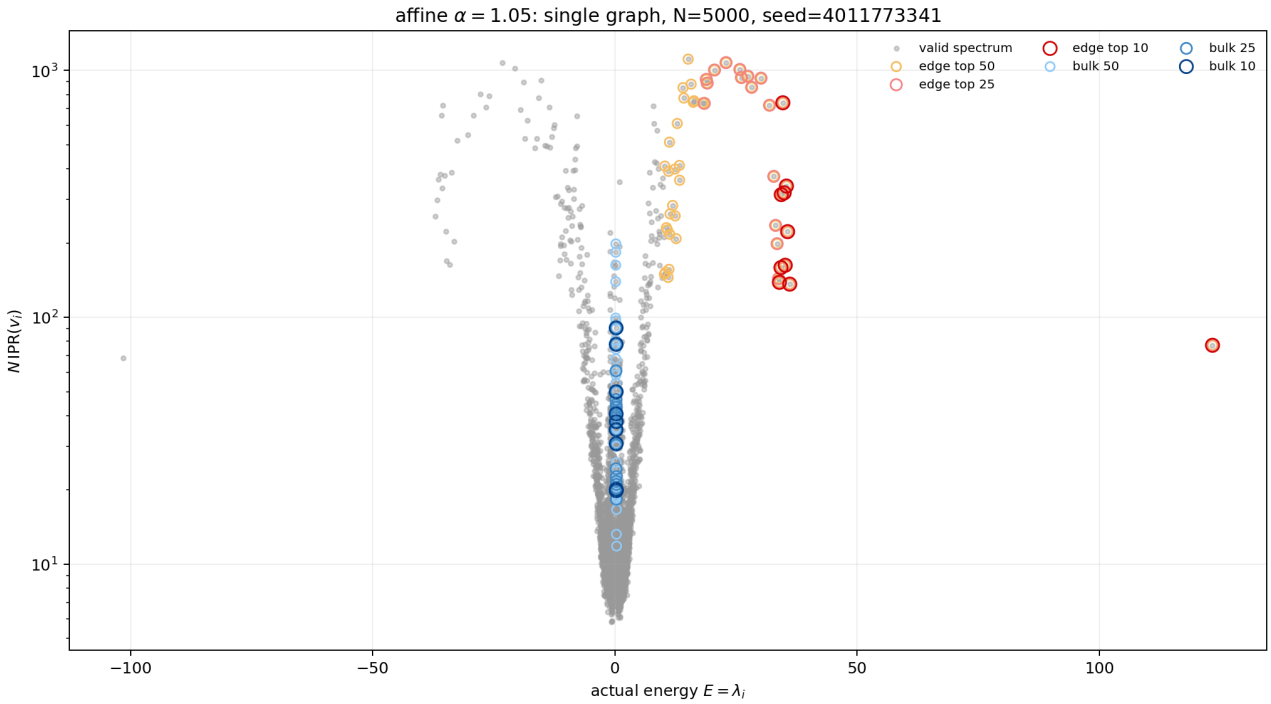


Figure 5: Original localization-versus-energy diagnostic for affine $\alpha = 1.05$.

Edge eigenpair microscope. Read the four panels together. Individual r_i values oscillate across almost the entire unit interval, so no single point or short run establishes Poisson or GOE statistics. By contrast, $N\,\mathrm{IPR}$ is already in the hundreds near the edge, rises above 10^3 around ranks 18–31, and remains far above its bulk scale. M_{10} stays near 0.45–0.50 through most of the top 50. The scatter panel shows that very large $N\,\mathrm{IPR}$ occurs at both small and large r_i; spacing ratio and localization strength are therefore not monotonically linked.

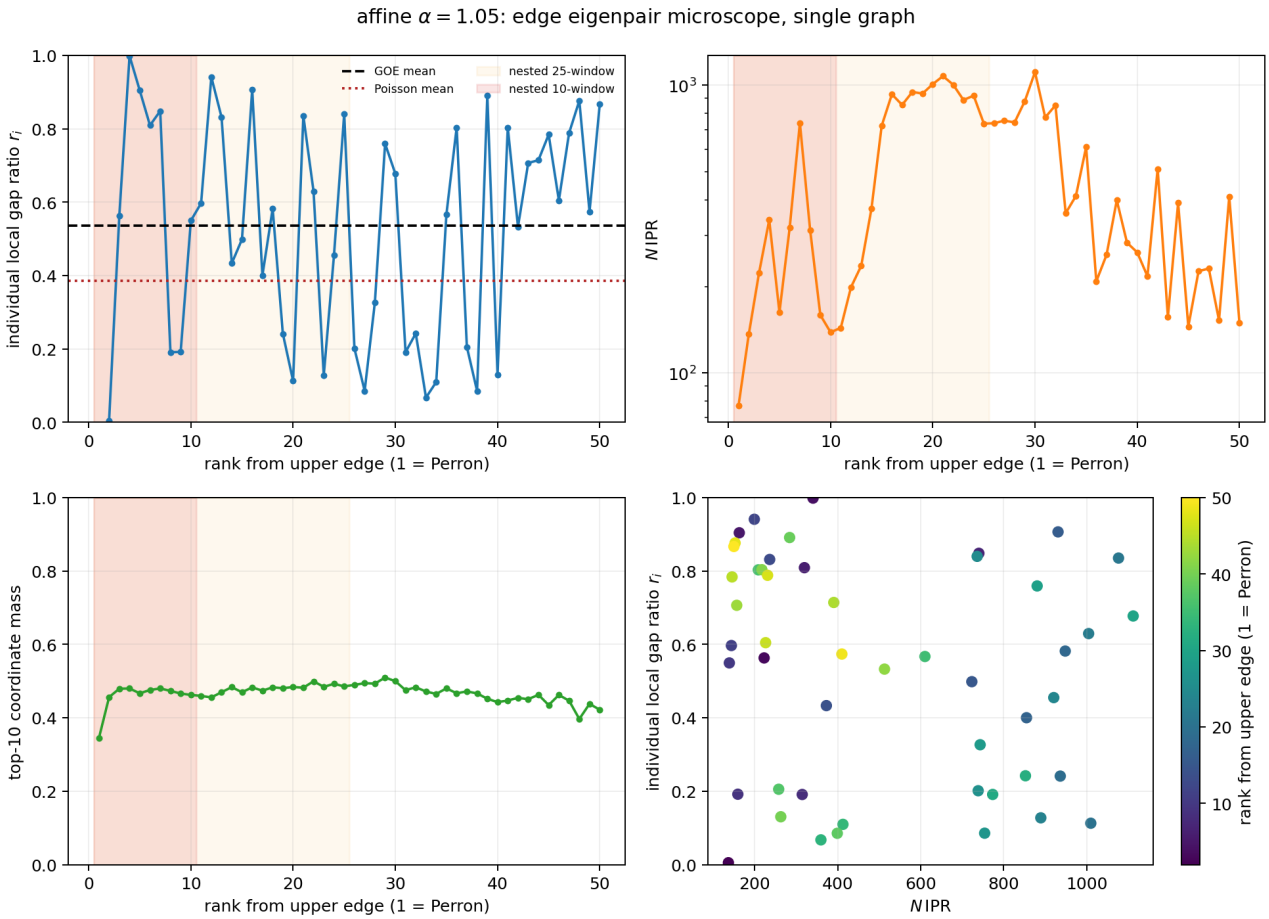


Figure 6: Original four-panel edge-rank eigenpair microscope for affine $\alpha = 1.05$.

Bulk eigenpair microscope. The centered zero-eigenvalue-free bulk still has irregular pointwise r_i, but its localization measures are much smaller: N IPR is mostly in the tens and M_{10} is generally about 0.1–0.3 near the selected center. The larger values at negative offsets show that the chosen 50-level display spans a localization gradient; the nested numerical tables are the controlled comparison.

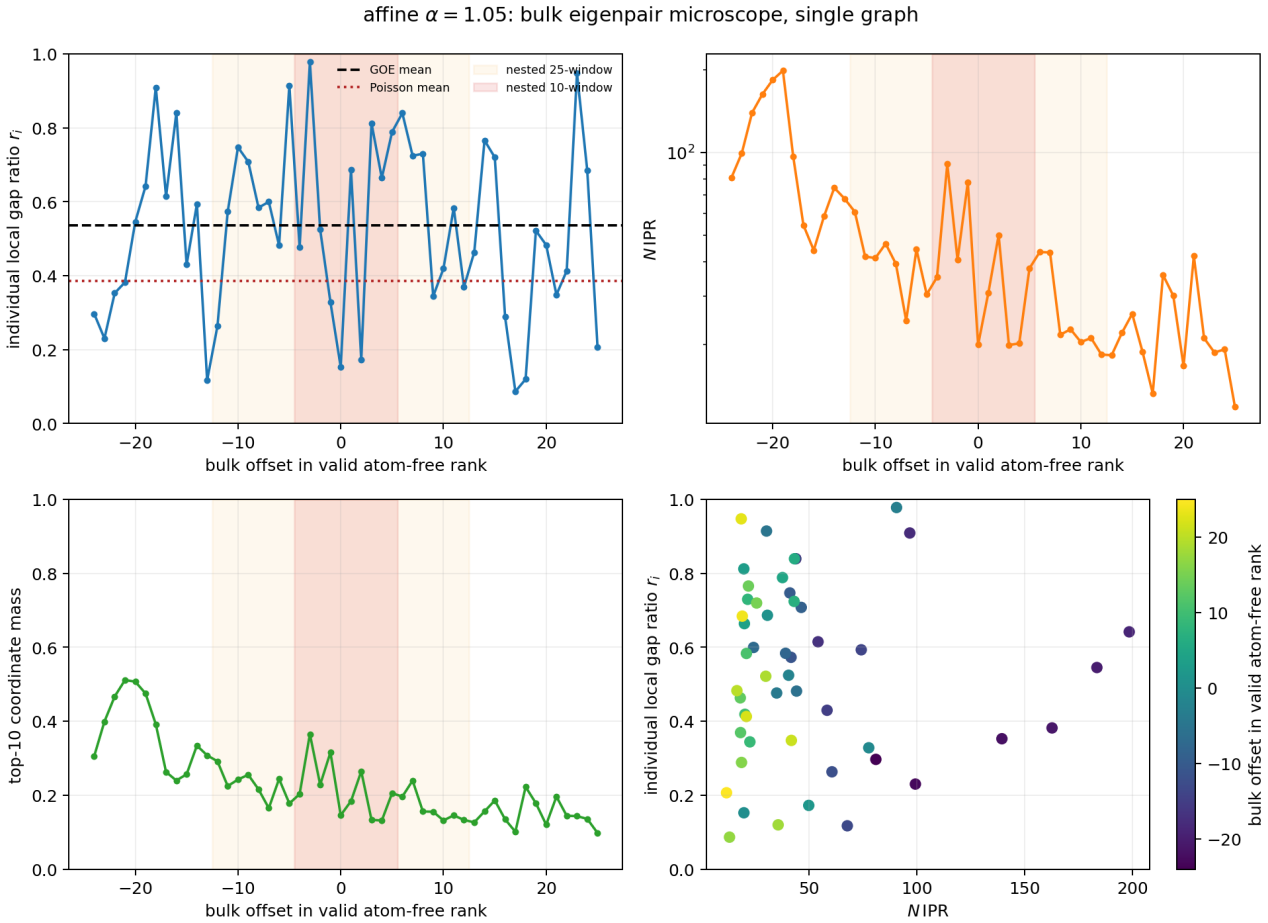


Figure 7: Original four-panel centered bulk-offset eigenpair microscope for affine $\alpha = 1.05$.

Direct edge-versus-bulk comparison. The boxplots separate edge from bulk most clearly in N IPR, M_{10}, top-support degree, induced density, and full-graph distance. Edge supports range from very dense hub cores to less dense deeper states, whereas bulk supports cluster near zero density. The edge distance z-scores are strongly negative, meaning substantially shorter paths than uniformly random ten-vertex sets.

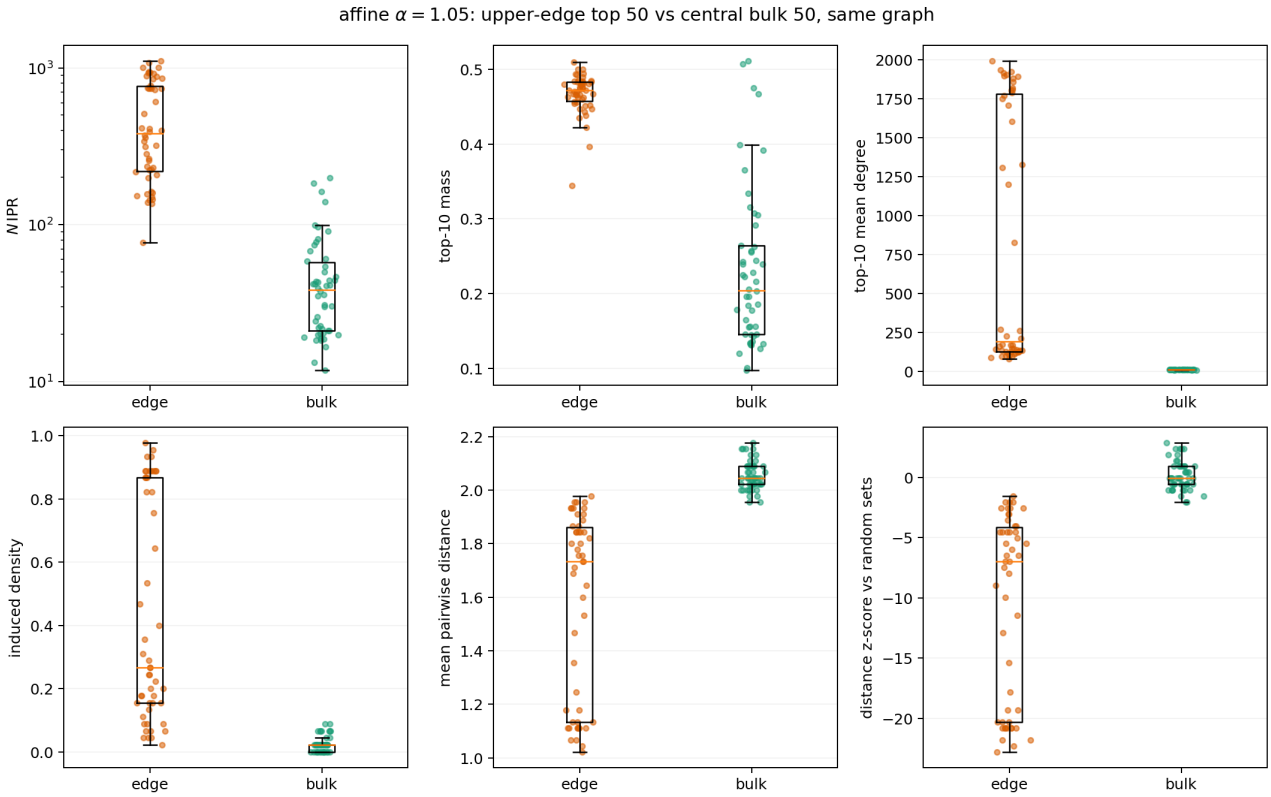


Figure 8: Original direct edge-versus-bulk comparison for affine $\alpha = 1.05$.

13.2 nonlinear $\zeta = 1.5$

Eigenvalue density. The full spectrum contains a smaller but still visible block of repeated zero eigenvalues. Removing those eigenvalues and the Perron root produces 4,759 continuous non-Perron levels. As in the affine stress case, the sharp central density peak makes raw gaps near zero small without implying localization on its own.

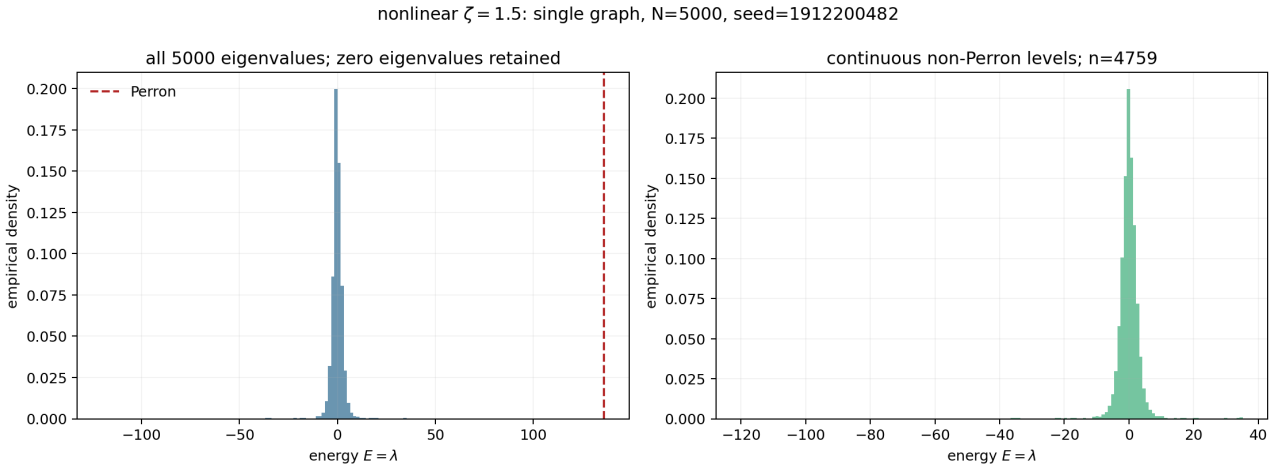


Figure 9: Original single-graph eigenvalue-density diagnostic for nonlinear $\zeta = 1.5$.

Raw and unfolded local spacing. Unfolding largely removes the energy-dependent gap scale. The well-populated interior panels are closer to the GOE shape than to the exponential Poisson reference; the outermost panels have only tens of gaps and should be read as descriptive. The full continuous-spectrum ratio mean is about 0.536, while the non-Perron top-10 strict mean is 0.387. That contrast is a narrow-edge effect, not a whole-spectrum Poisson regime.

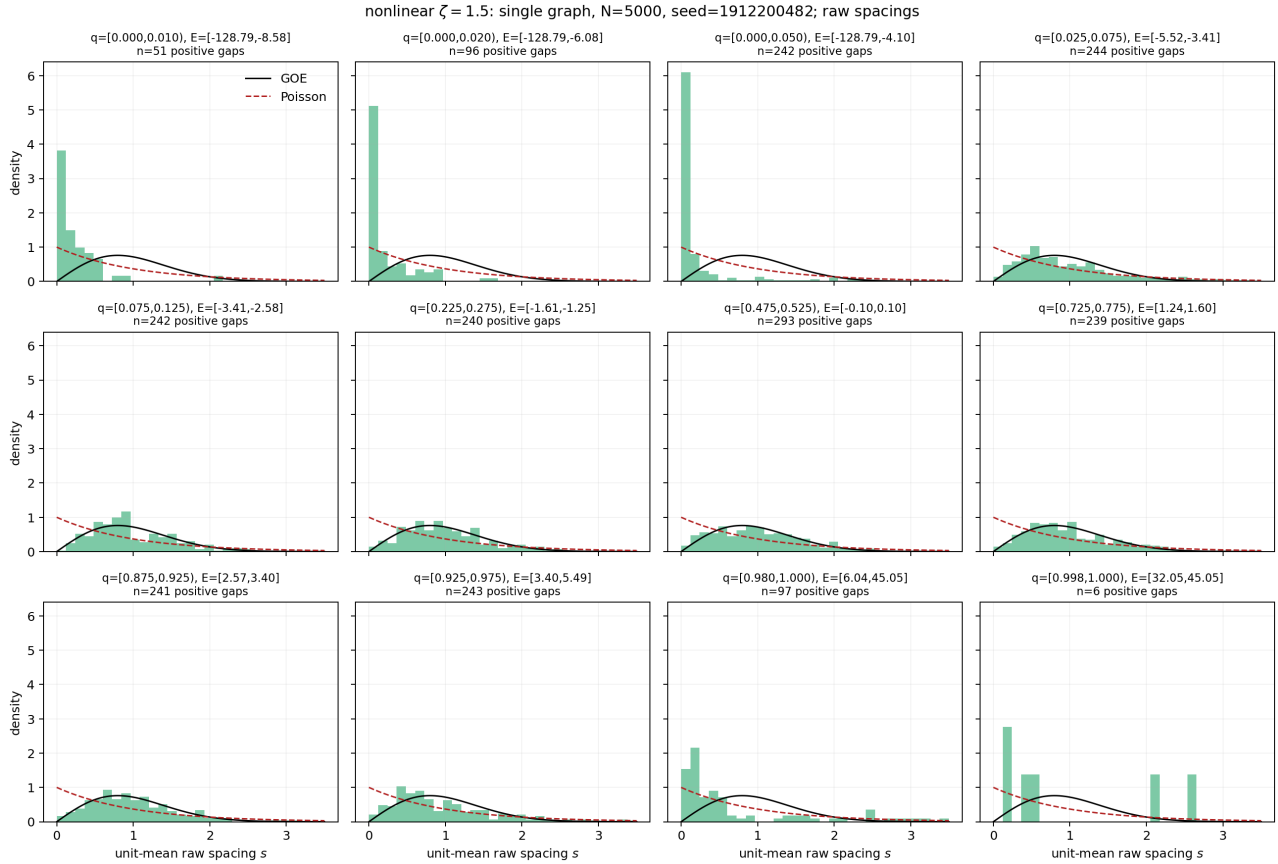


Figure 10: Original local raw-spacing histograms for nonlinear $\zeta = 1.5$.

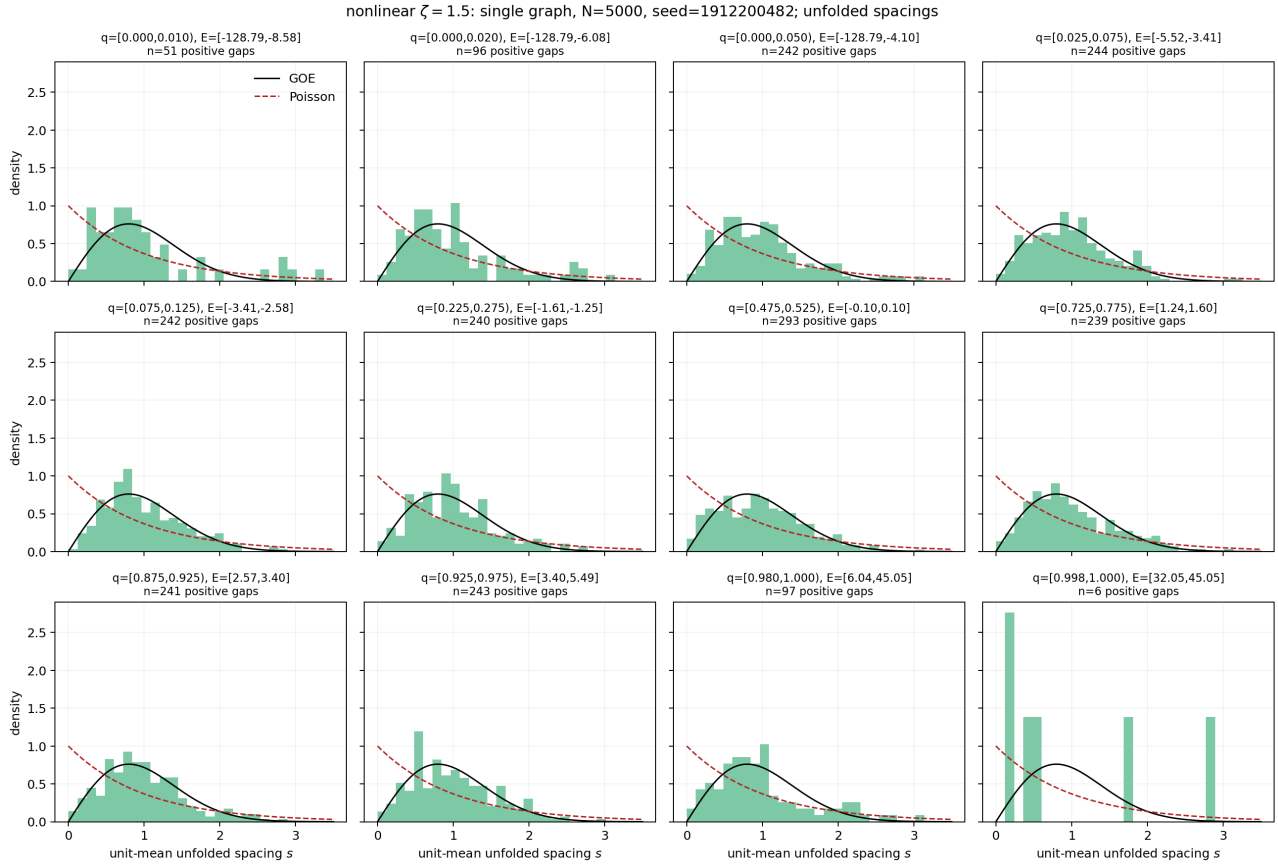


Figure 11: Original local unfolded-spacing histograms for nonlinear $\zeta = 1.5$.

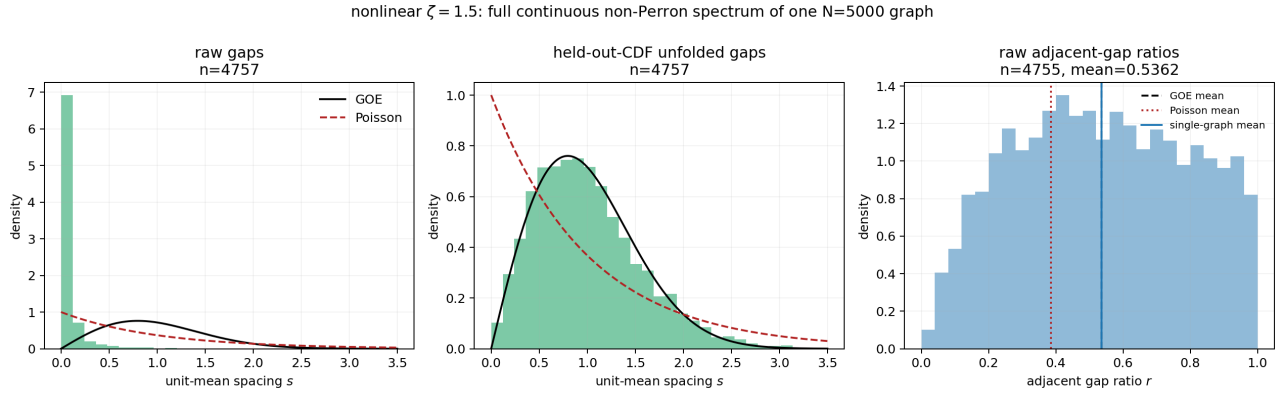


Figure 12: Original full continuous-spectrum spacing and ratio diagnostic for nonlinear $\zeta = 1.5$.

Localization versus energy. The continuous bulk near $E = 0$ lies around $N\,\mathrm{IPR} \approx 3$–$15$, whereas the positive edge rises to the hundreds and occasionally near 10^3. The positive-edge rings are cleanly separated vertically from the bulk rings, making eigenvector localization the strongest visual edge signal.

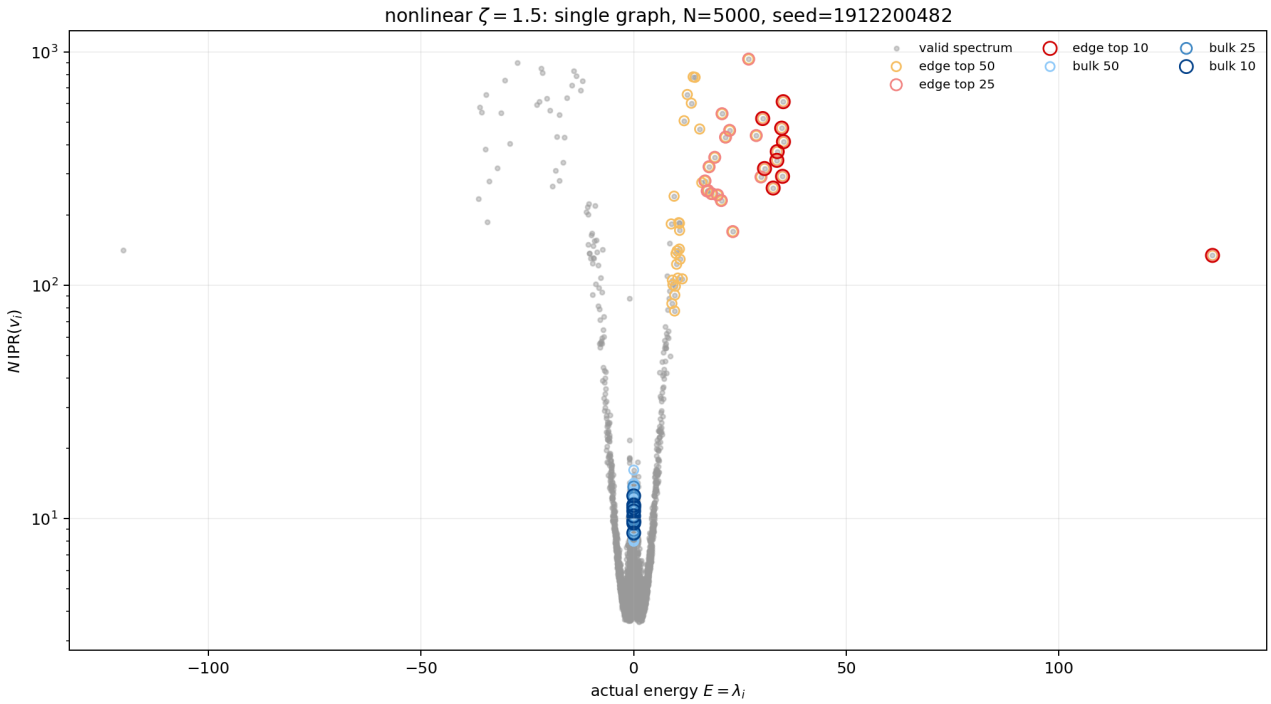


Figure 13: Original localization-versus-energy diagnostic for nonlinear $\zeta = 1.5$.

Edge eigenpair microscope. This is the four-panel figure highlighted in the request. The upper-left points are individual $r_i = \min(s_{i-1}, s_i)/\max(s_{i-1}, s_i)$ values; their large scatter is expected. Rank 2 is near zero because one adjacent gap reaches toward the isolated Perron root, which is why the strict non-Perron table begins its usable ratios at rank 3. The top-10 strict mean is near Poisson, but the top-25 and top-50 means move upward. Meanwhile N IPR stays in the hundreds through roughly rank 30 and then drops toward about 100; M_{10} stays near 0.48 through roughly rank 30 before declining. The scatter panel again has no monotone r_i–localization relation.

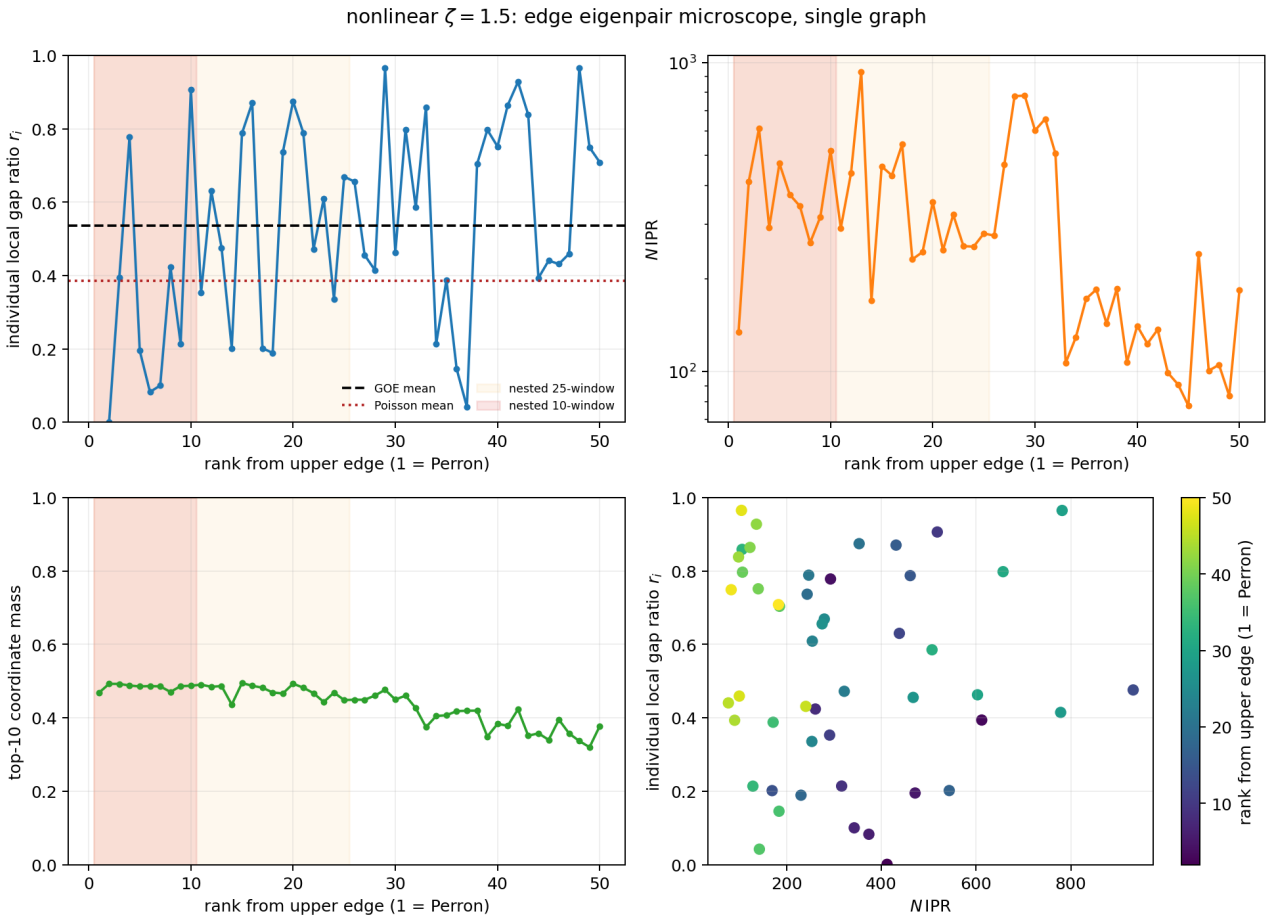


Figure 14: Original four-panel edge-rank eigenpair microscope for nonlinear $\zeta = 1.5$.

Bulk eigenpair microscope. Around the selected zero-eigenvalue-free center, N IPR remains tightly confined near 8–16 and M_{10} near 0.06–0.10, despite large fluctuations in individual r_i. This is the cleanest example of why the bulk can be extended even when a short sequence of ratios happens to contain both Poisson-like and GOE-like values.

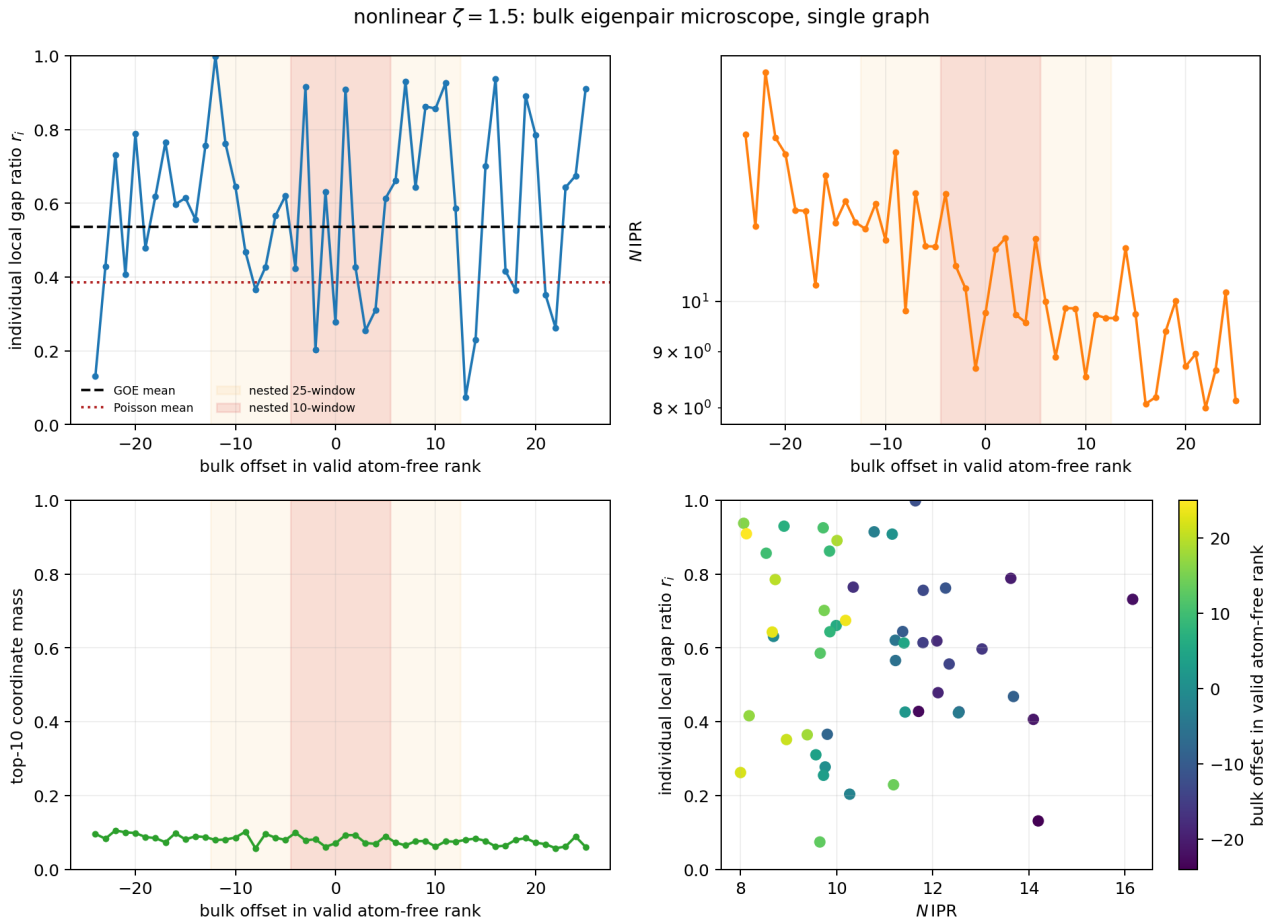


Figure 15: Original four-panel centered bulk-offset eigenpair microscope for nonlinear $\zeta = 1.5$.

Direct edge-versus-bulk comparison. The edge and bulk distributions are almost disjoint in N IPR and M_{10}. Edge top-mass vertices also have much higher degree, form substantially denser induced subgraphs, and have shorter full-graph paths. The bulk induced density is exactly zero for these displayed supports, while the edge density is often large; this is a structural localization signature independent of the ambiguous ratio average.

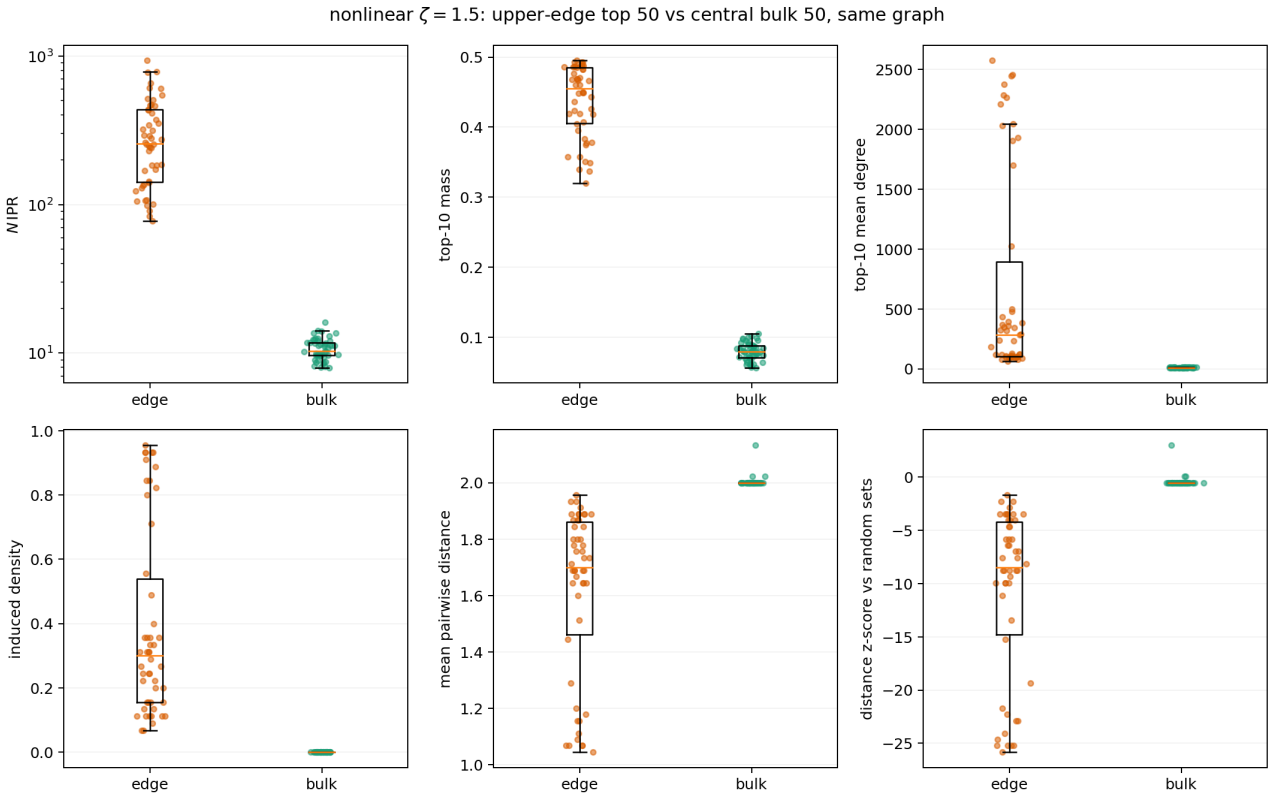


Figure 16: Original direct edge-versus-bulk comparison for nonlinear $\zeta = 1.5$.

13.3 affine $\alpha = 2.0$

Eigenvalue density. This model has no repeated zero eigenvalues in the stored graph: 4,999 continuous non-Perron levels remain. Its broad, smooth density makes it the clean control for separating edge localization from zero-eigenvalue or rank-deficiency effects.

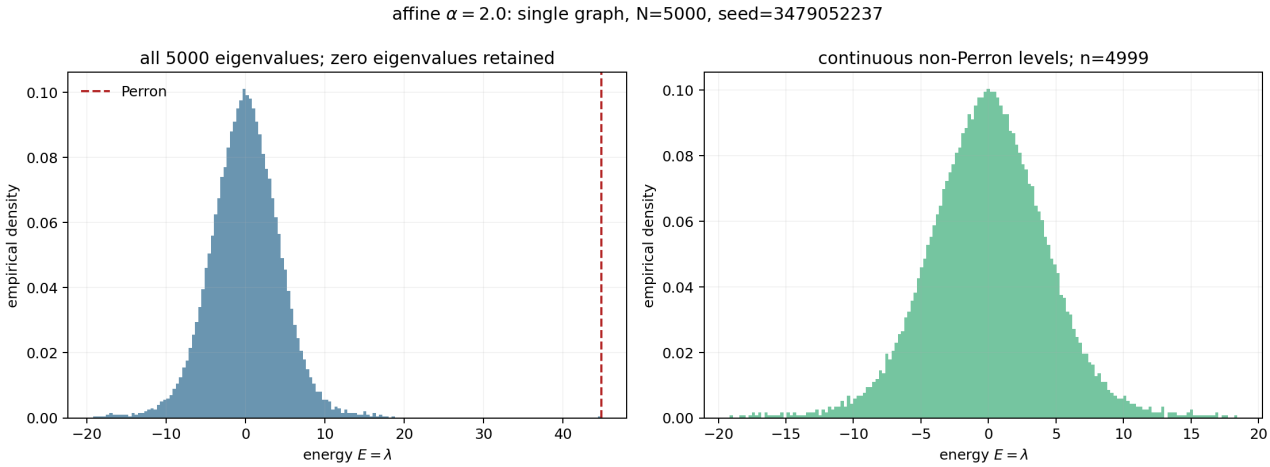


Figure 17: Original single-graph eigenvalue-density diagnostic for affine $\alpha = 2.0$.

Raw and unfolded local spacing. The unfolded interior histograms track the GOE reference reasonably well, while extreme-edge panels have far fewer gaps and visible discreteness. The full-spectrum ratio mean is about 0.525, again close to GOE. Thus localization at this edge does not require an overall Poisson gap-ratio law in this single graph.

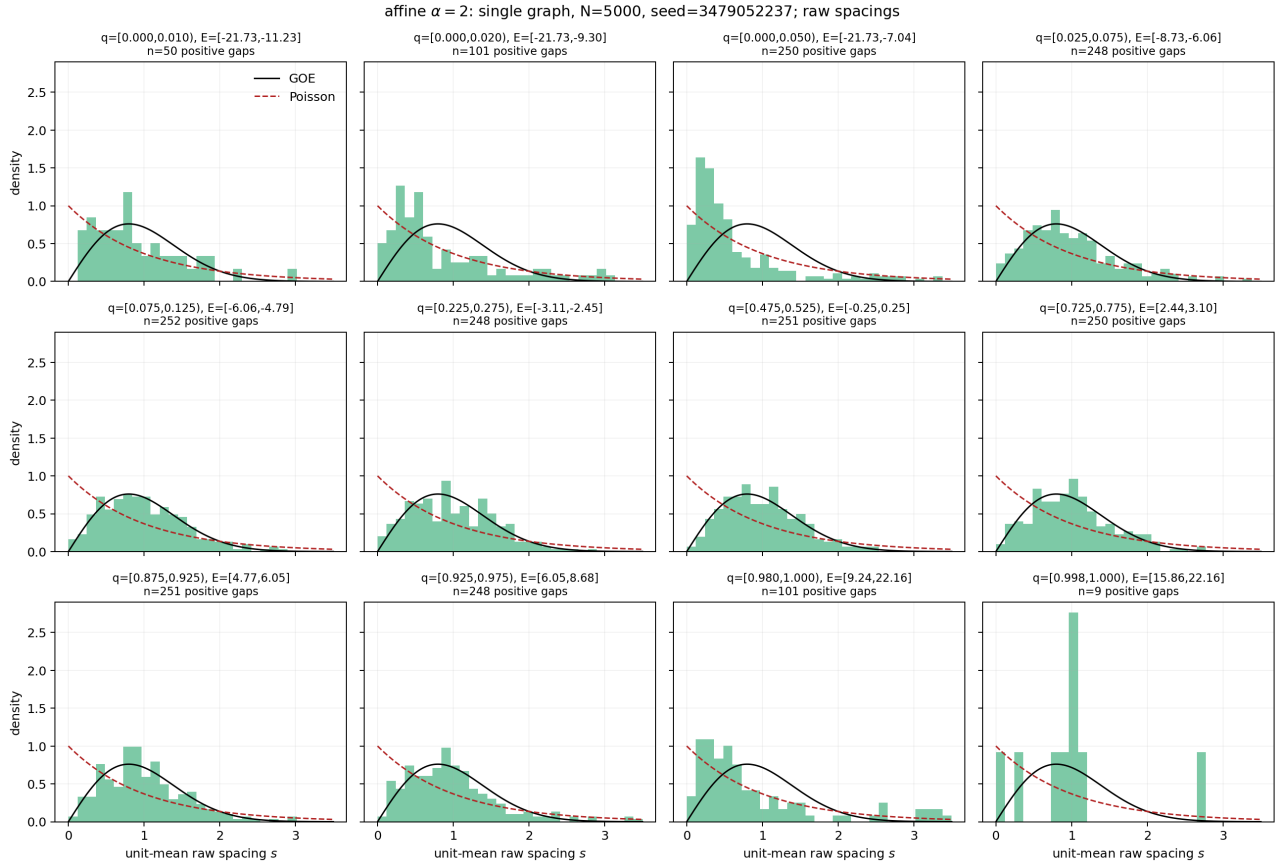


Figure 18: Original local raw-spacing histograms for affine $\alpha = 2.0$.

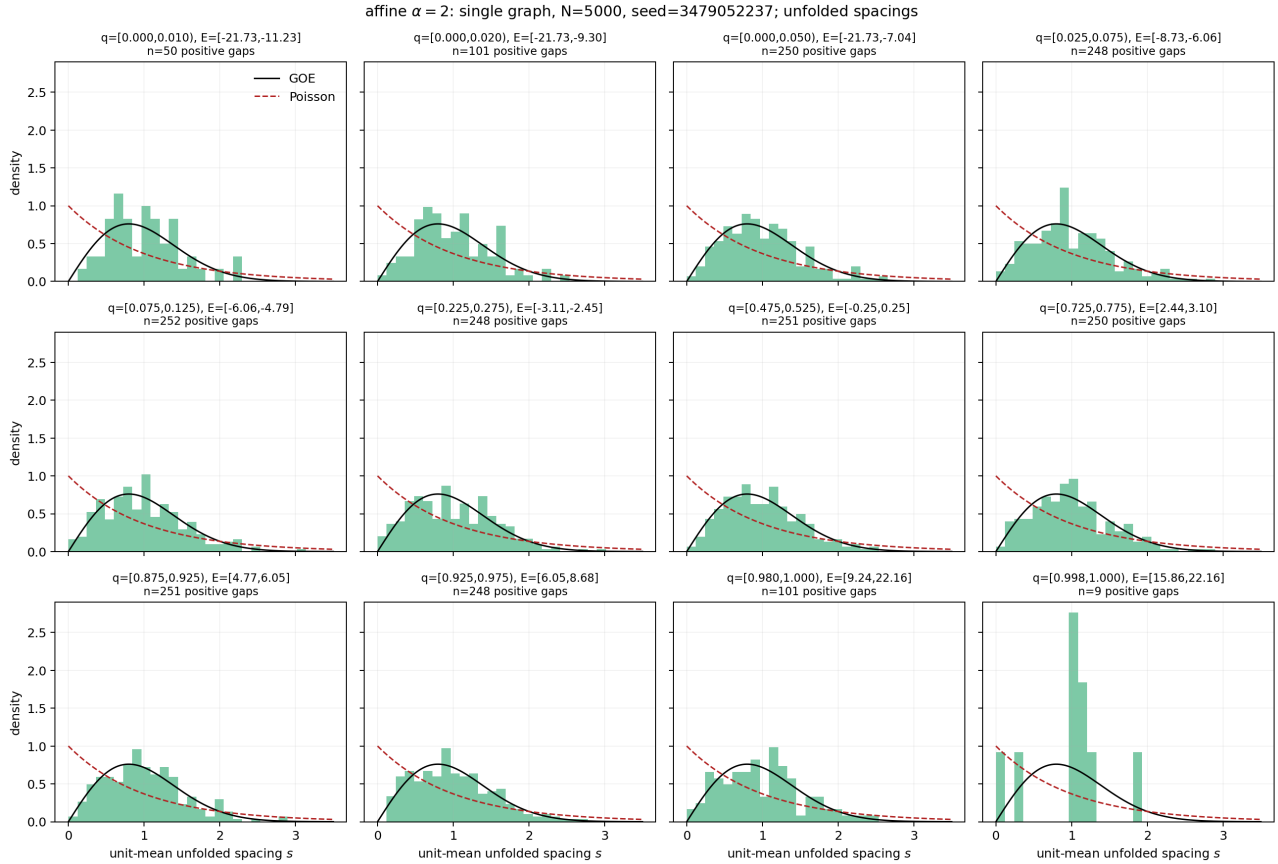


Figure 19: Original local unfolded-spacing histograms for affine $\alpha = 2.0$.

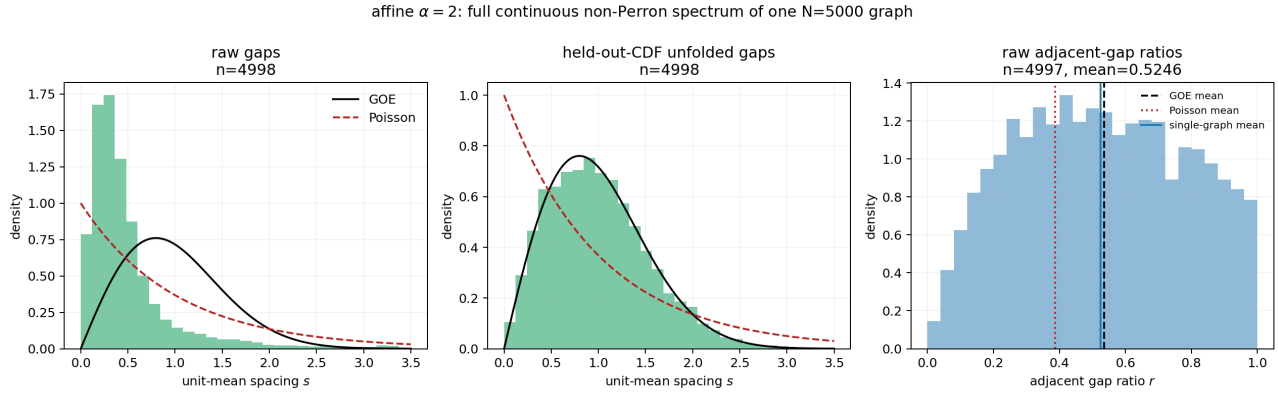


Figure 20: Original full continuous-spectrum spacing and ratio diagnostic for affine $\alpha = 2.0$.

Localization versus energy. The N IPR bowl is smooth and symmetric apart from the isolated Perron point: values are near 4 in the bulk and rise into the tens or hundreds toward the tails. This provides the clearest visual mobility-edge-like crossover of the three graphs, although one realization cannot locate a sharp transition.

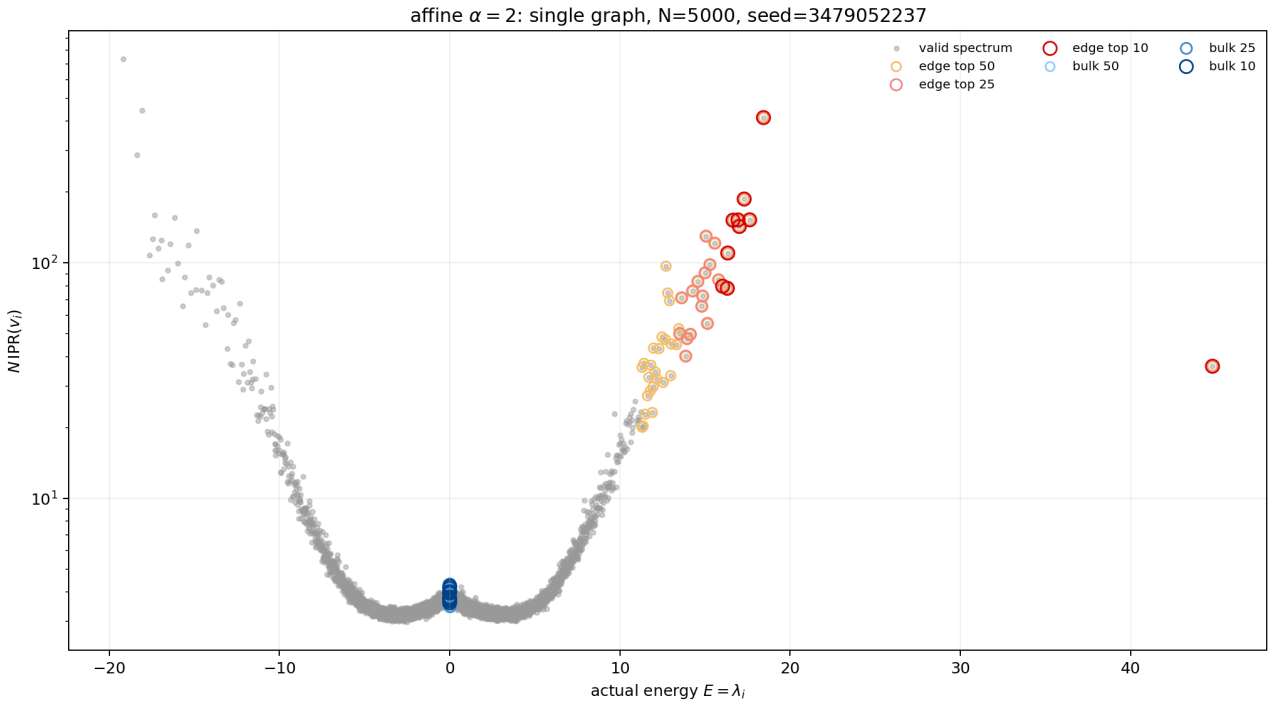


Figure 21: Original localization-versus-energy diagnostic for affine $\alpha = 2.0$.

Edge eigenpair microscope. The upper-right and lower-left panels decline fairly steadily with rank: $N\,\mathrm{IPR}$ falls from hundreds to a few tens, and M_{10} falls from roughly 0.4 to 0.15–0.2. Individual ratios still jump between small and large values, and the scatter plot shows no single-valued relation between r_i and $N\,\mathrm{IPR}$. The localization crossover is much clearer than any crossover in r_i.

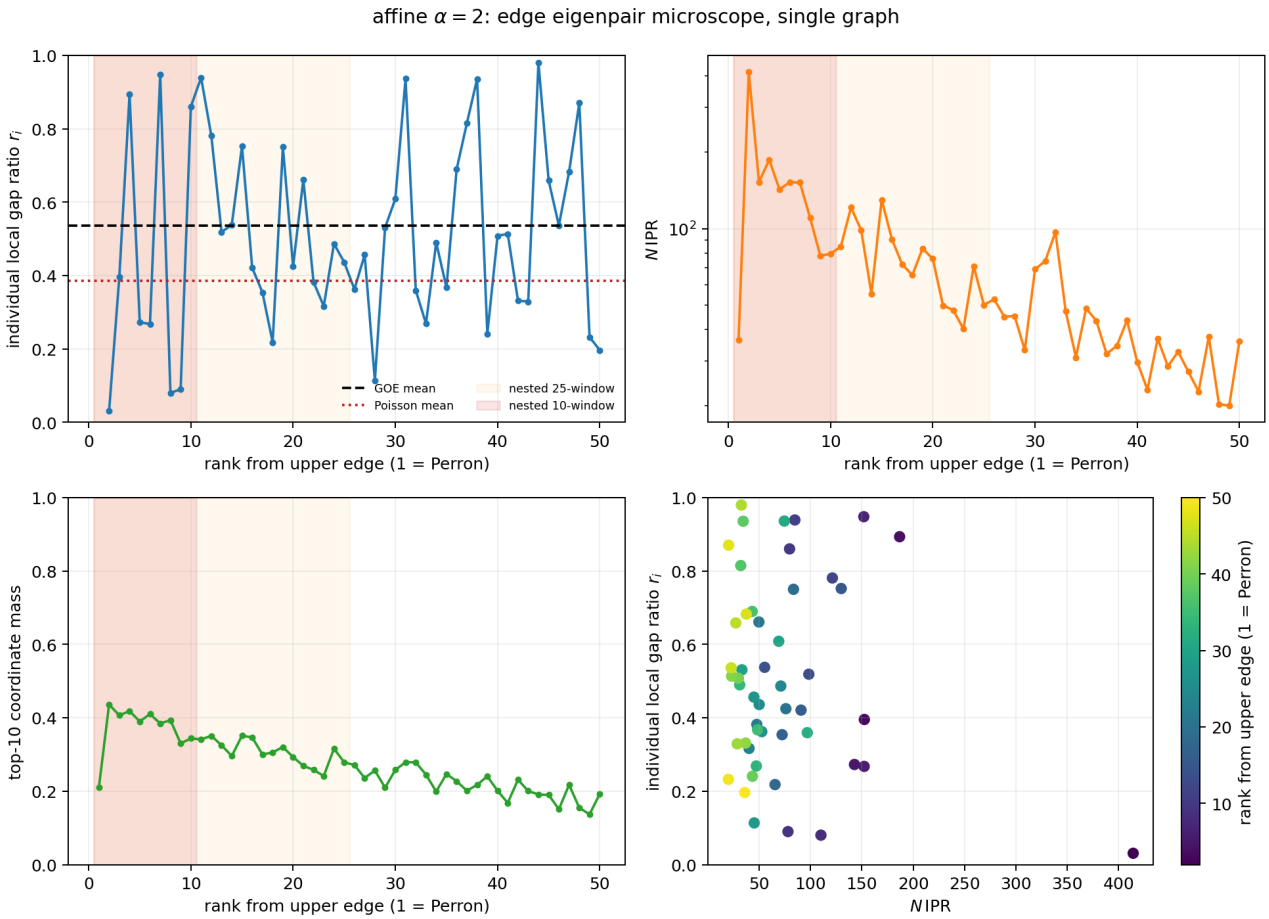


Figure 22: Original four-panel edge-rank eigenpair microscope for affine $\alpha = 2.0$.

Bulk eigenpair microscope. The bulk localization panels are nearly flat: N IPR stays around 3.5–4.3 and M_{10} around 0.03. Pointwise gap ratios remain noisy, but their fluctuations occur on an eigenvector background that is uniformly extended relative to the edge.

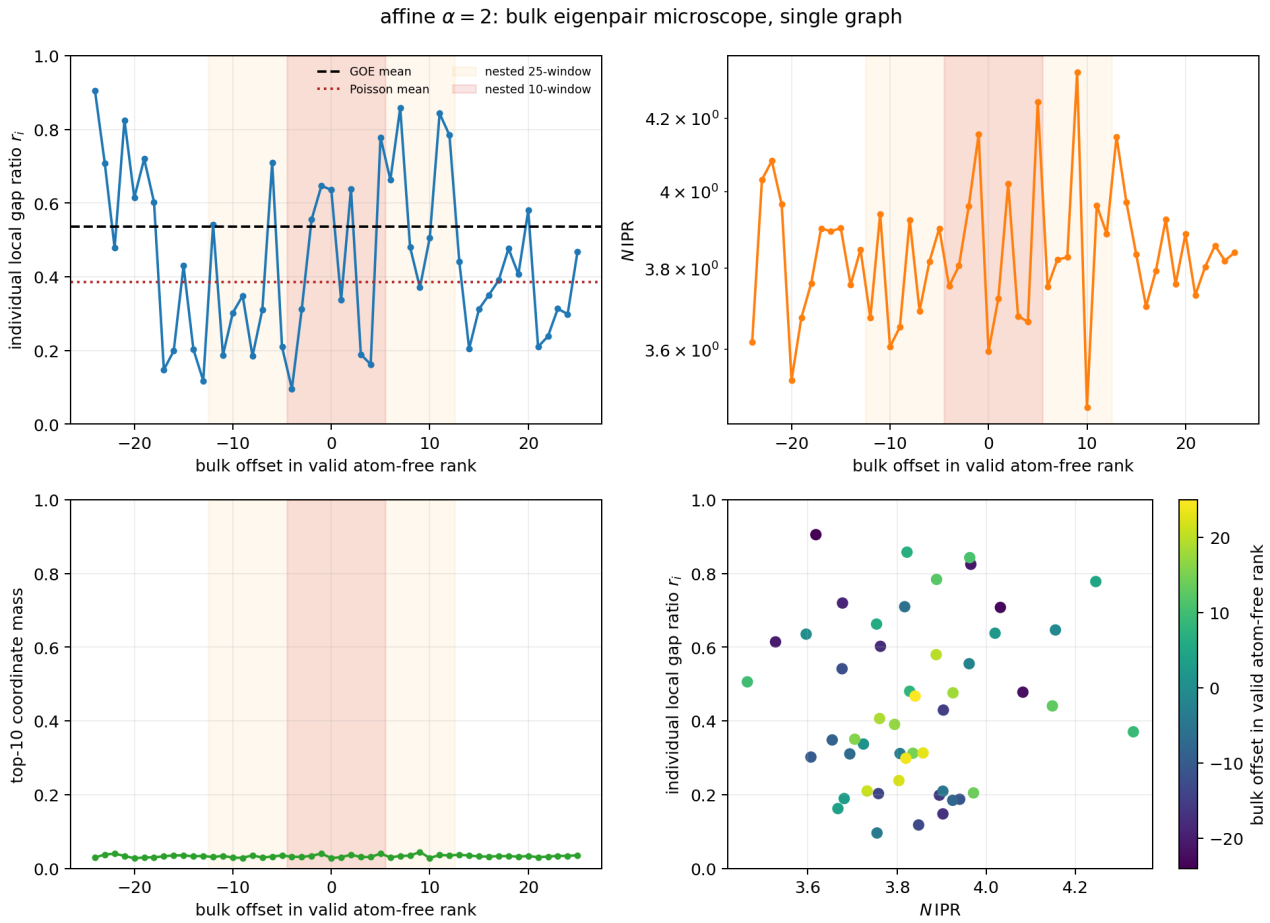


Figure 23: Original four-panel centered bulk-offset eigenpair microscope for affine $\alpha = 2.0$.

Direct edge-versus-bulk comparison. Every eigenvector/support metric separates edge and bulk: edge N IPR and M_{10} are much larger, its selected vertices have degrees in the hundreds rather than about ten, induced density is substantial rather than zero, and pairwise distance is roughly 1–2 rather than about 3. The strongly negative distance z-scores confirm compactness relative to uniform random supports.

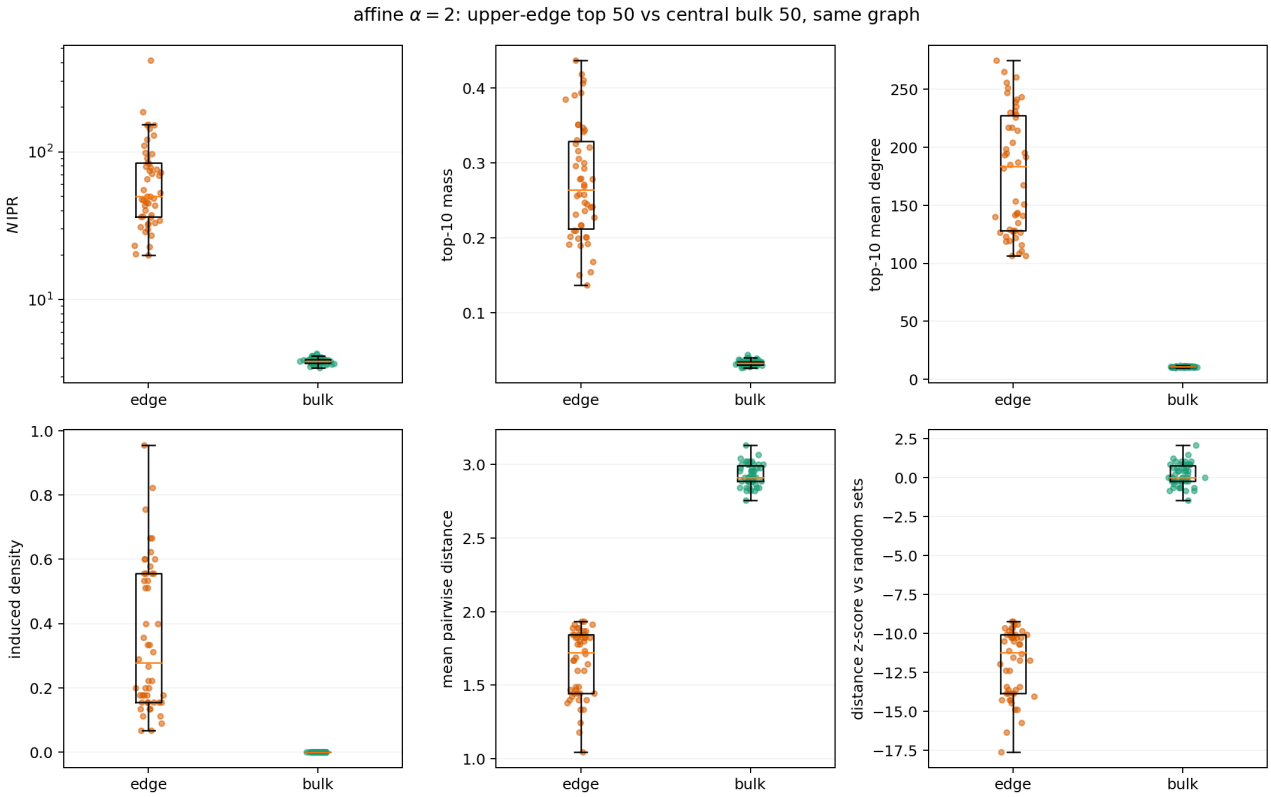
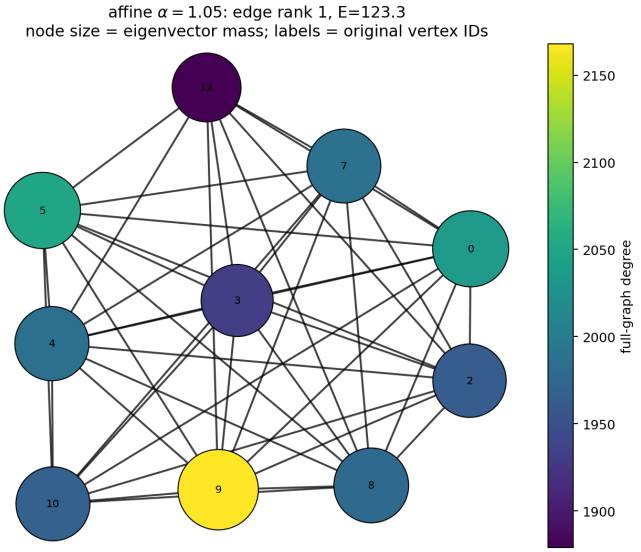


Figure 24: Original direct edge-versus-bulk comparison for affine $\alpha = 2.0$.

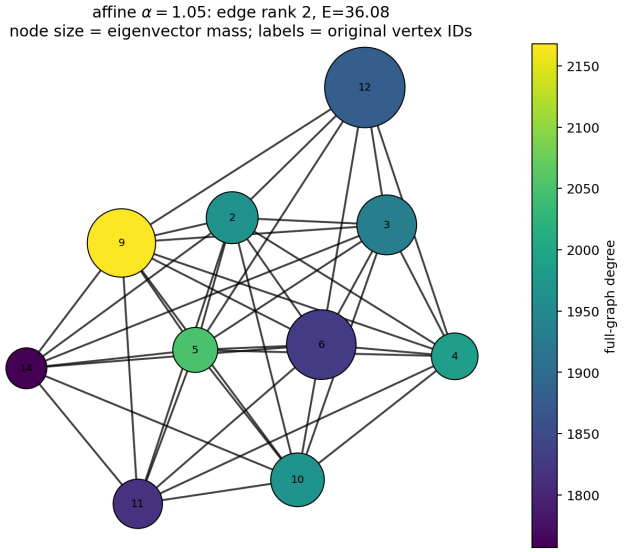
14 Original induced top-mass support graphs

These are the original test drawings: edge ranks 1–10 include the Perron state at rank 1, and the bulk panels are the centered zero-eigenvalue-free bulk-10 window. They are retained exactly as recorded; no replacement or newly generated induced graph is included. Coordinate tables remain omitted. In every panel, the number printed inside a node is its original vertex ID in the full graph, while node area is proportional to that vertex's eigenvector mass $|v_j|^2$; node color, not node size, encodes its full-graph degree. Across the three models, the edge drawings visually agree with the numerical density, component, LCC, degree, and distance summaries: the selected vertices usually form one dense hub-rich component. The bulk drawings are almost edgeless, often ten singleton components. This does not contradict full-graph distances near two or three because paths through vertices outside the displayed ten are not drawn.

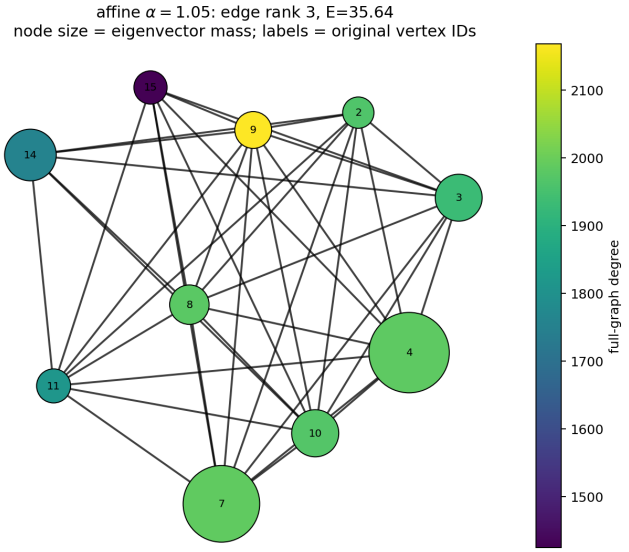
14.1 affine $\alpha = 1.05$: original edge ranks 1–10



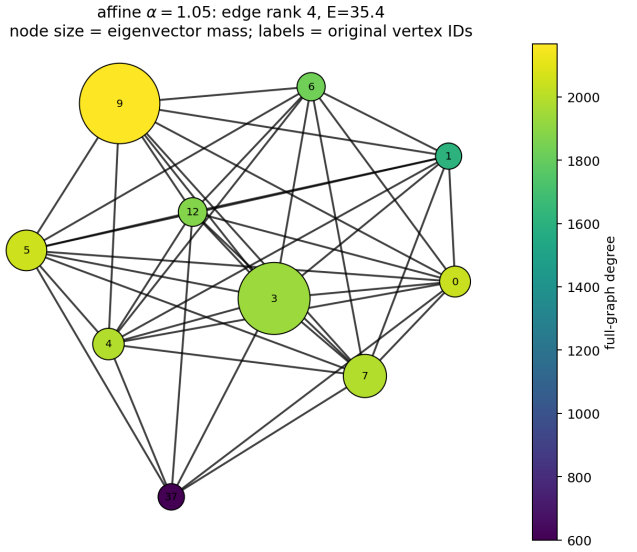
edge rank 001



edge rank 002

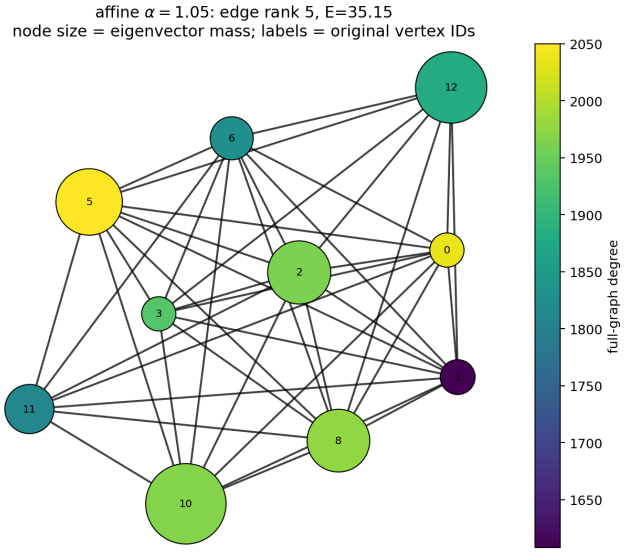


edge rank 003

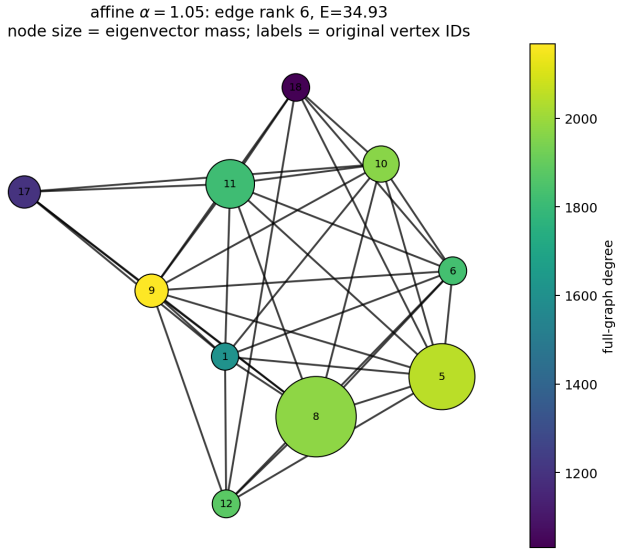


edge rank 004

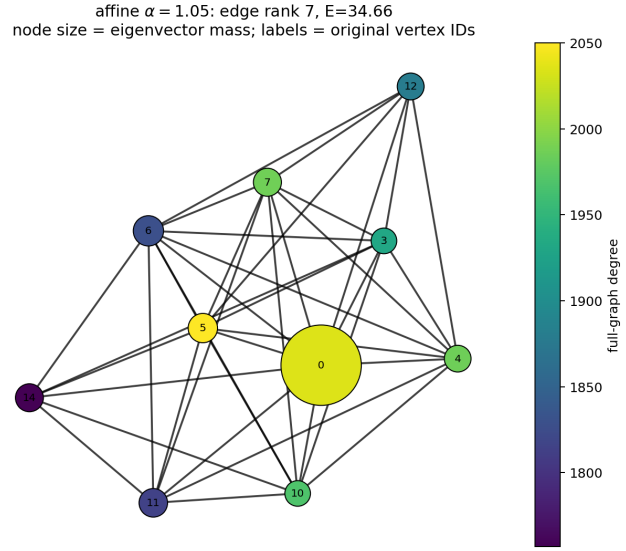
Figure 25: Original edge induced supports, ranks 1–4.



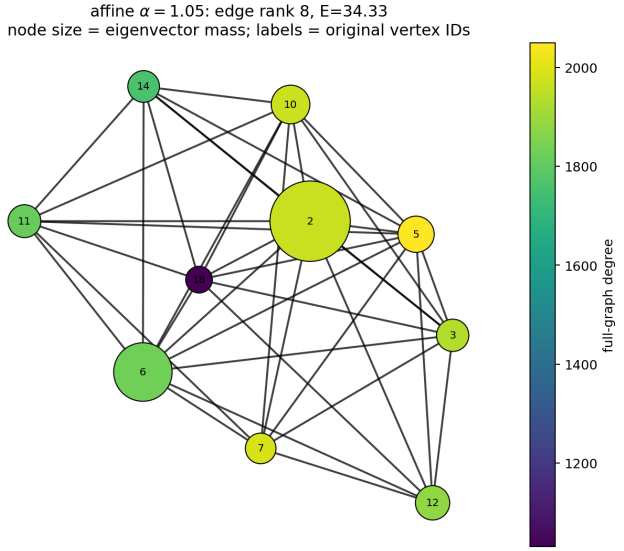
edge rank 005



edge rank 006

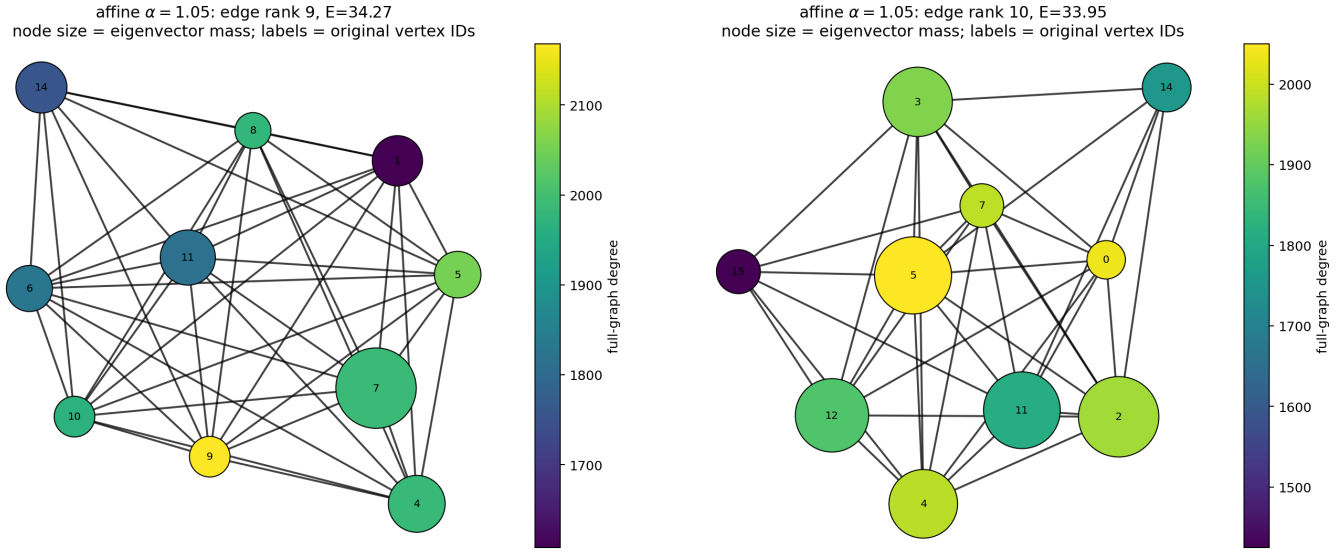


edge rank 007



edge rank 008

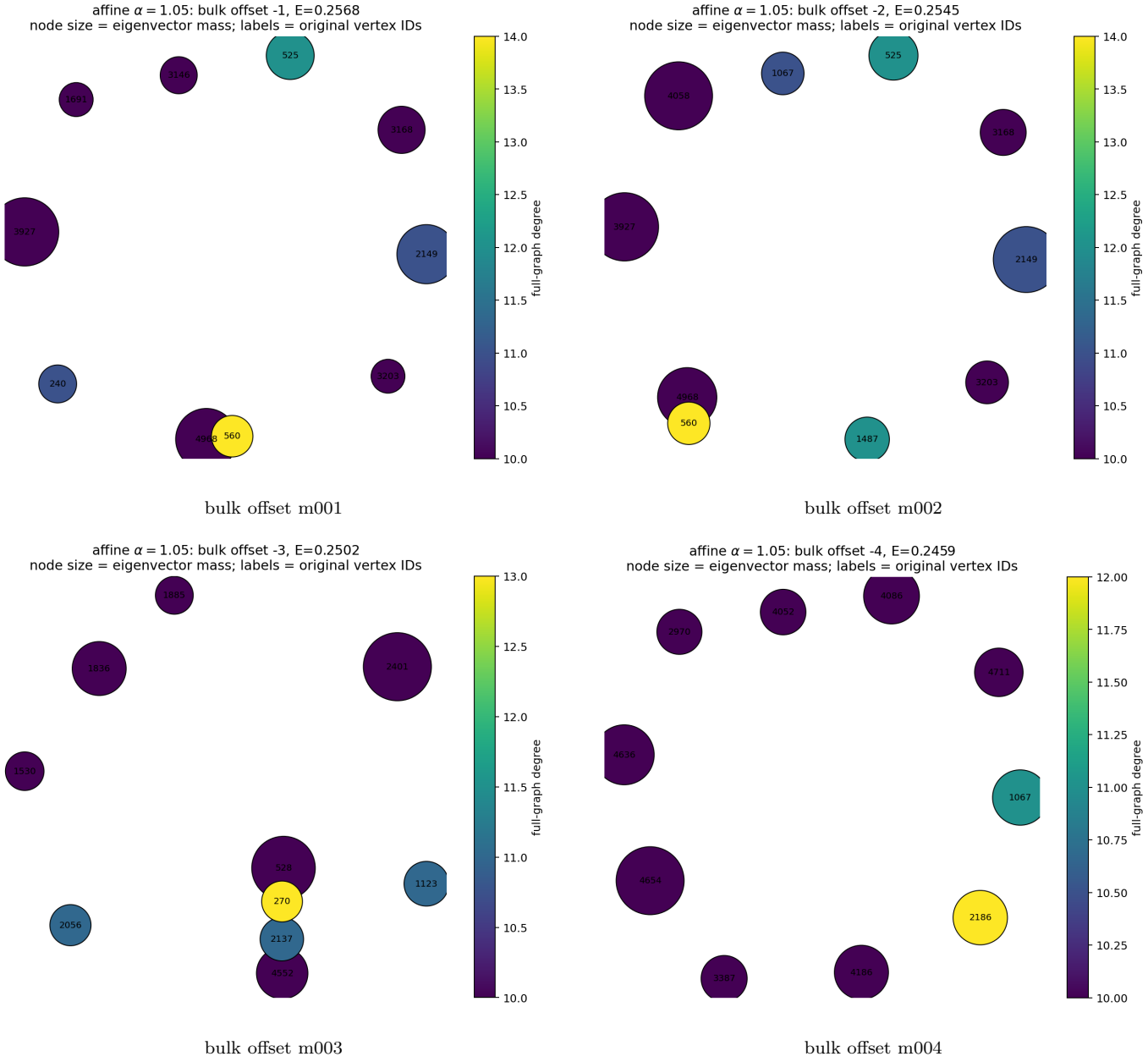
Figure 26: Original edge induced supports, ranks 5–8.



edge rank 009 edge rank 010

Figure 27: Original edge induced supports, ranks 9–10.

14.2 affine $\alpha = 1.05$: original centered bulk window



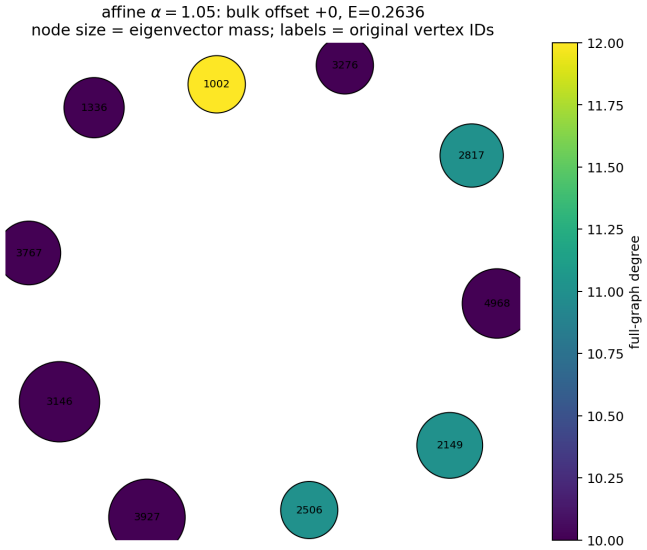
bulk offset m001

bulk offset m002

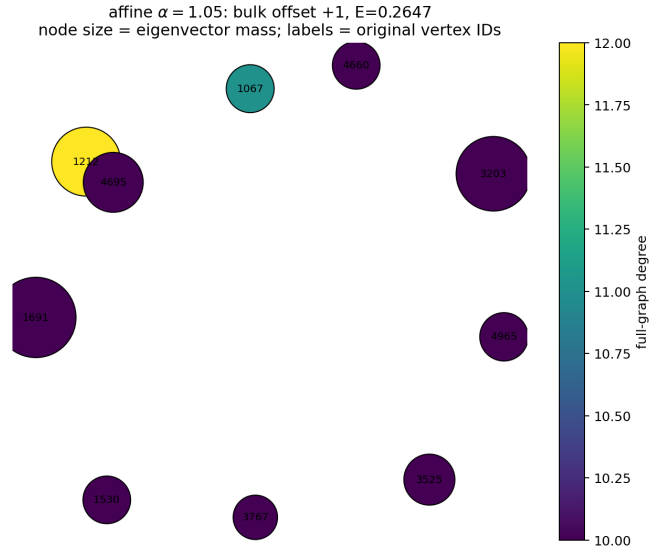
bulk offset m003

bulk offset m004

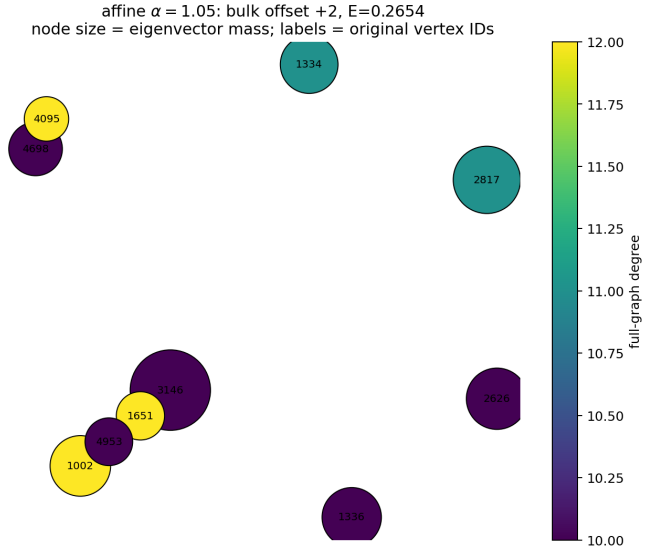
Figure 28: Original centered bulk induced supports.



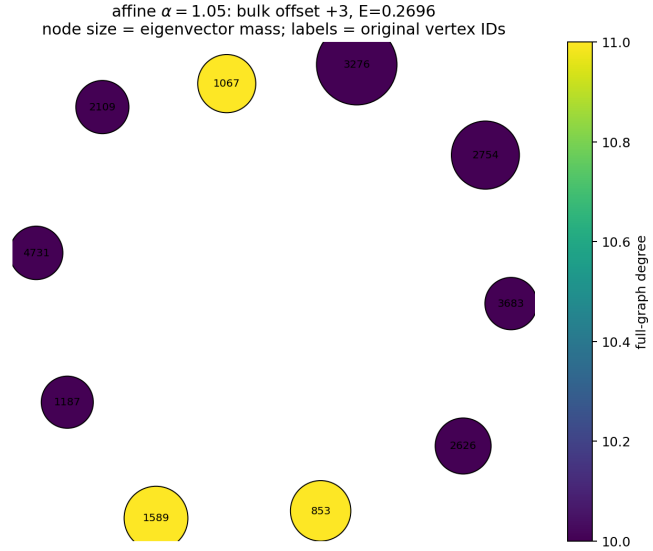
bulk offset p000



bulk offset p001

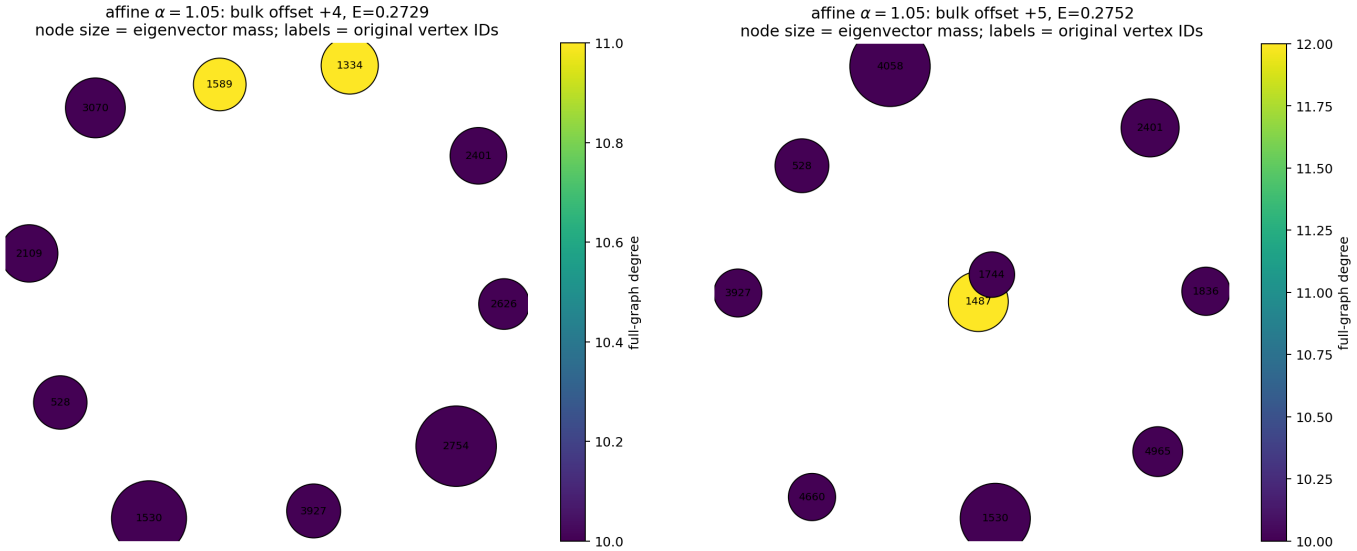


bulk offset p002



bulk offset p003

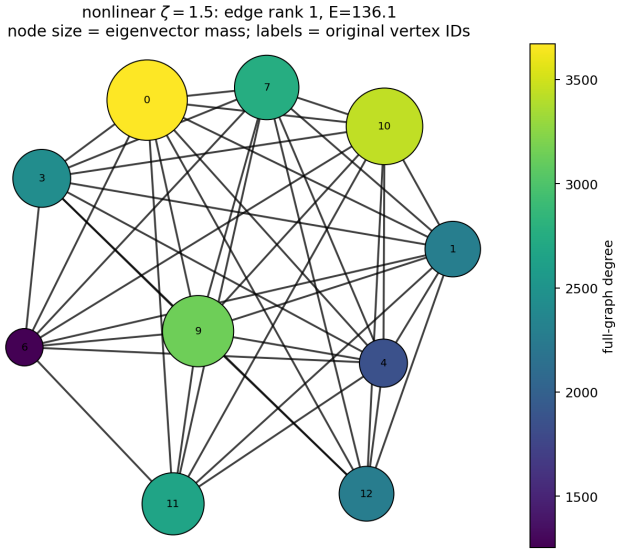
Figure 29: Original centered bulk induced supports.



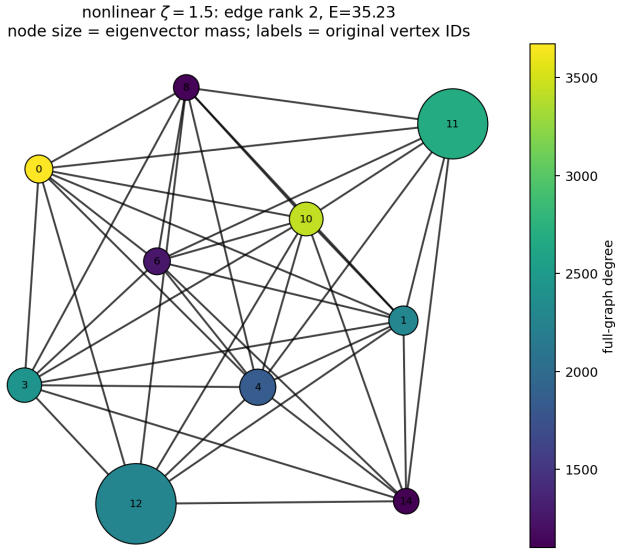
bulk offset p004 bulk offset p005

Figure 30: Original centered bulk induced supports.

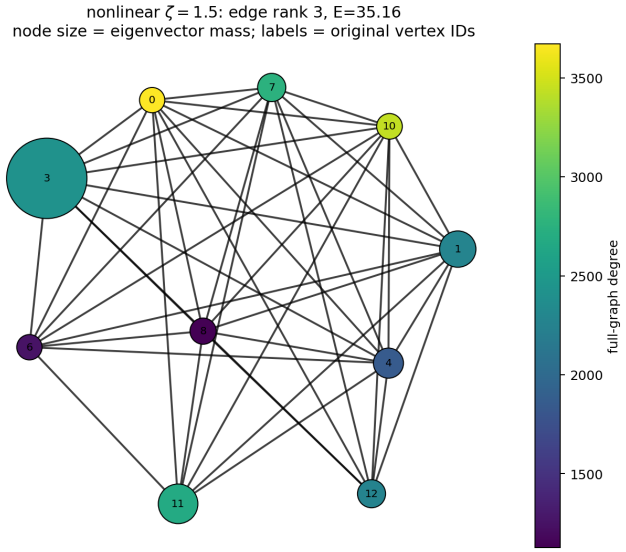
14.3 nonlinear $\zeta = 1.5$: original edge ranks 1–10



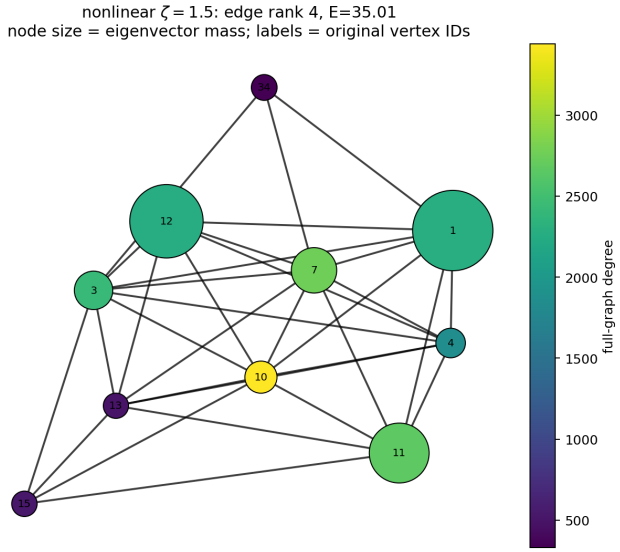
edge rank 001



edge rank 002

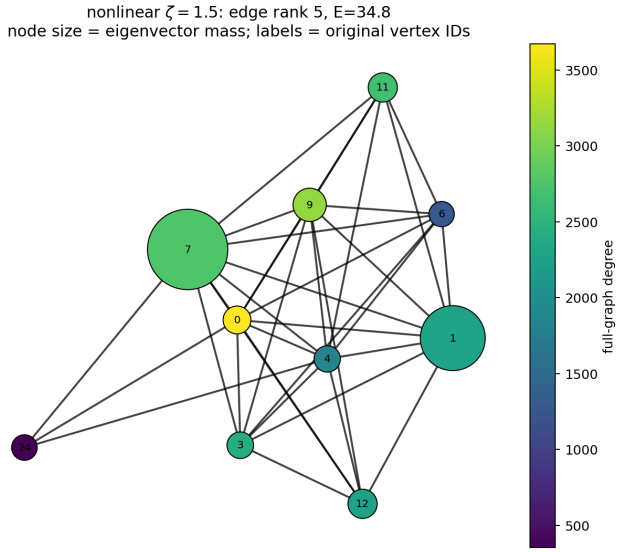


edge rank 003

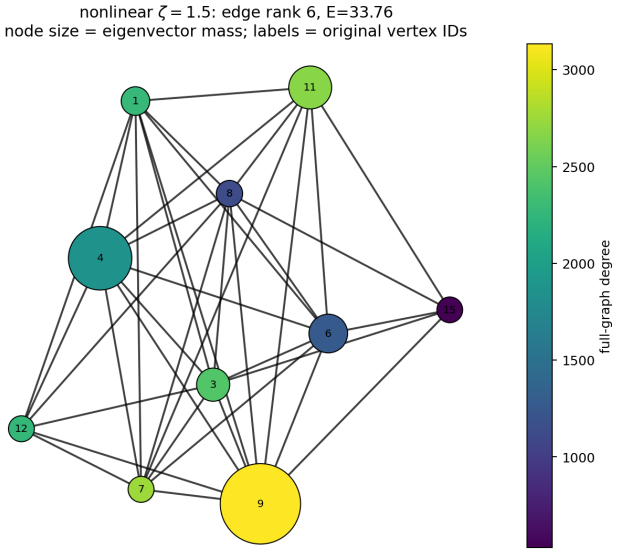


edge rank 004

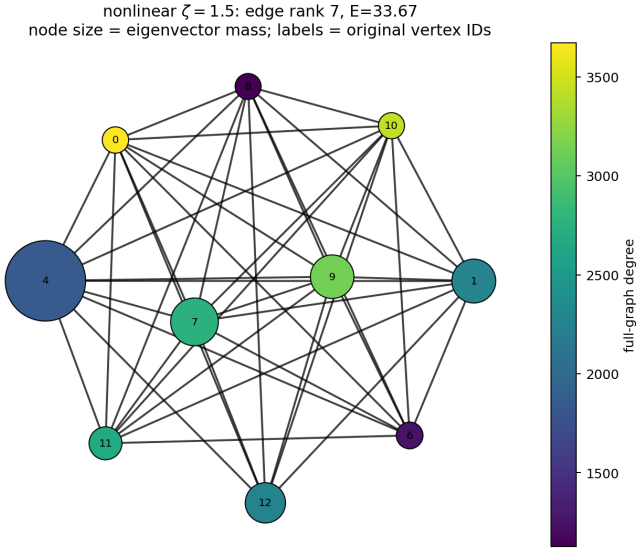
Figure 31: Original edge induced supports, ranks 1–4.



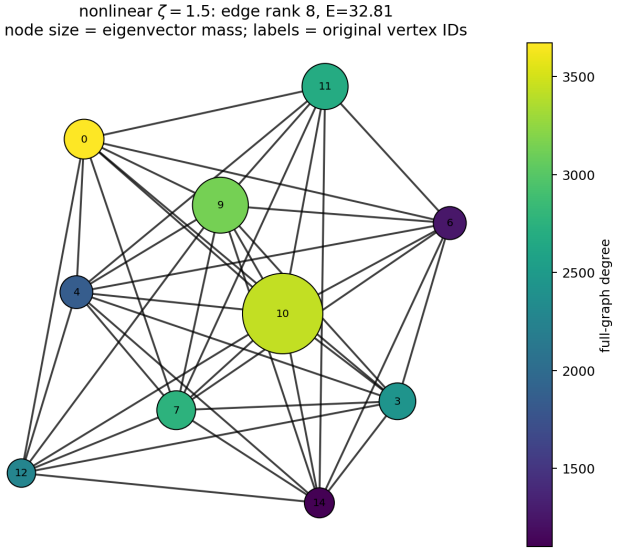
edge rank 005



edge rank 006

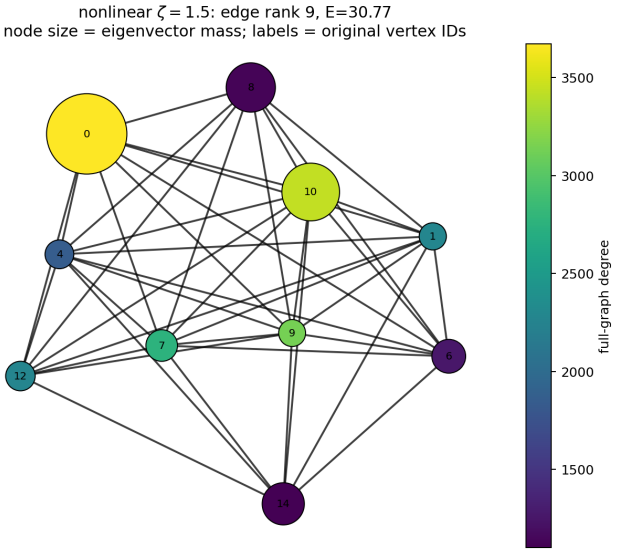


edge rank 007

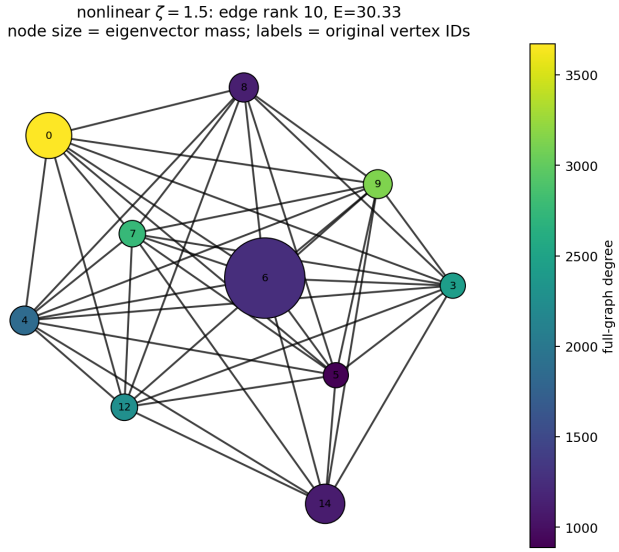


edge rank 008

Figure 32: Original edge induced supports, ranks 5–8.



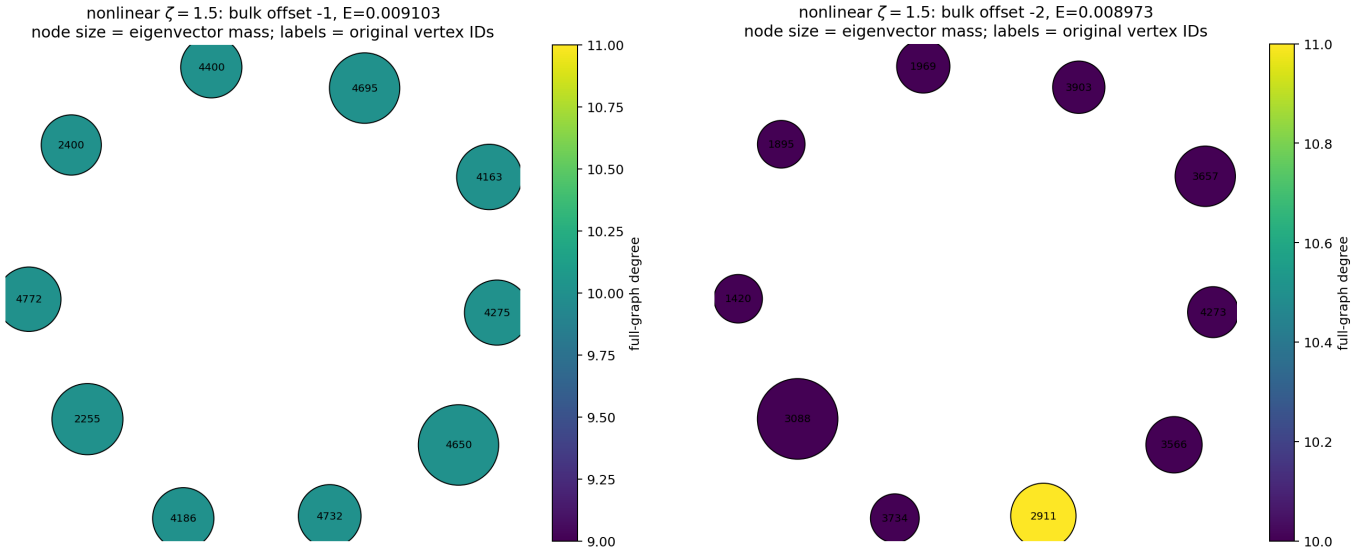
edge rank 009



edge rank 010

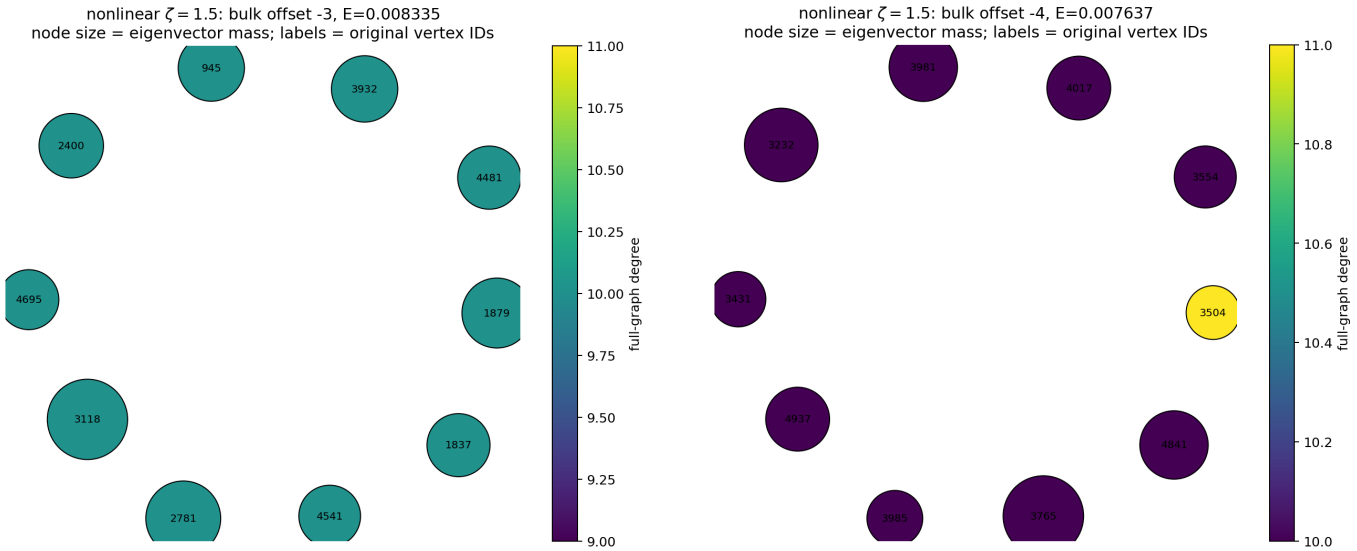
Figure 33: Original edge induced supports, ranks 9–10.

14.4 nonlinear $\zeta = 1.5$: original centered bulk window



bulk offset m001

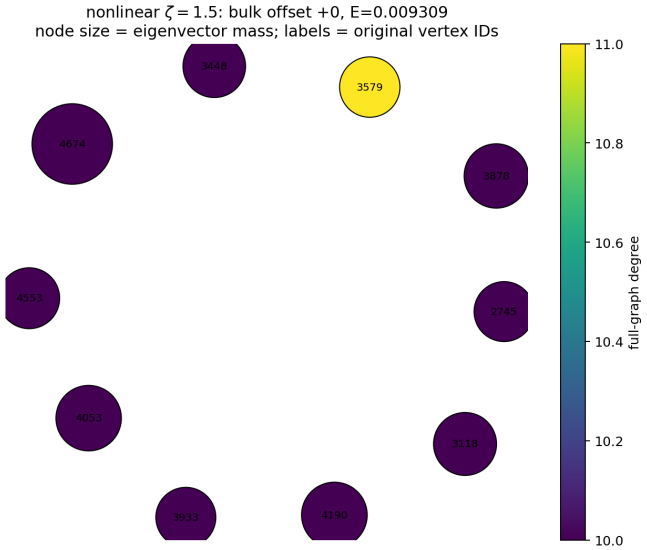
bulk offset m002



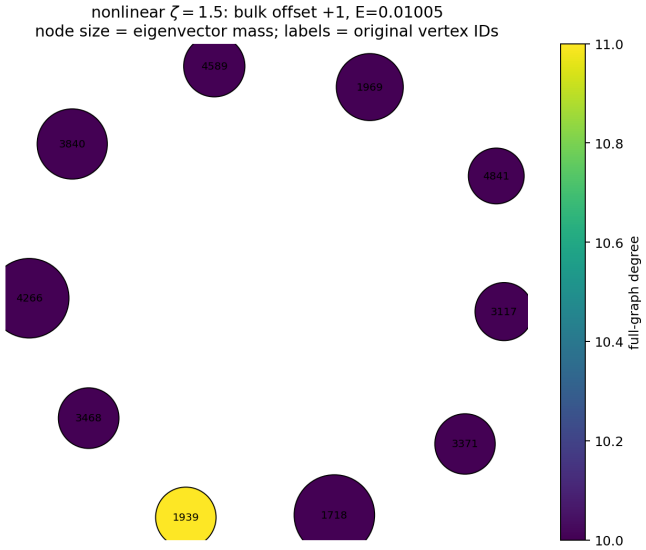
bulk offset m003

bulk offset m004

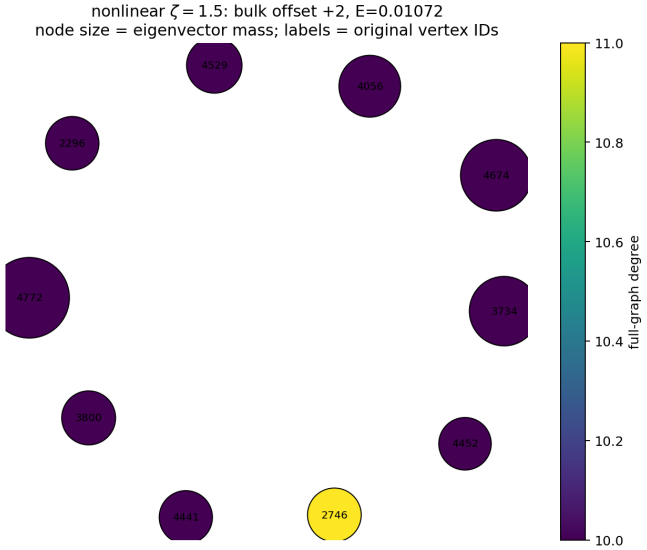
Figure 34: Original centered bulk induced supports.



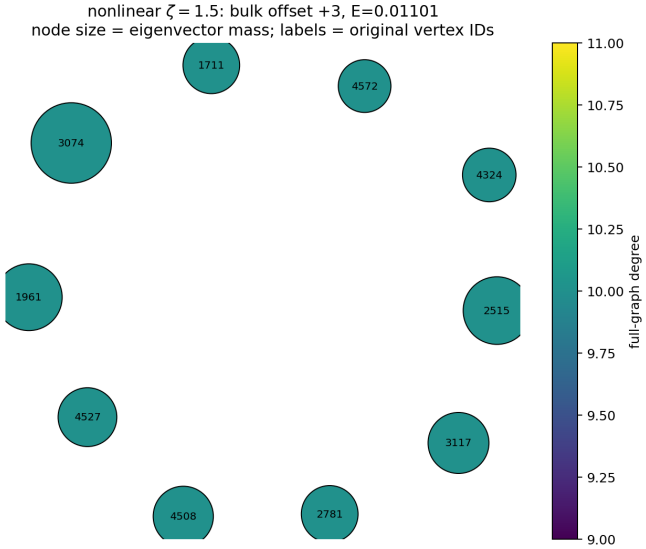
bulk offset p000



bulk offset p001

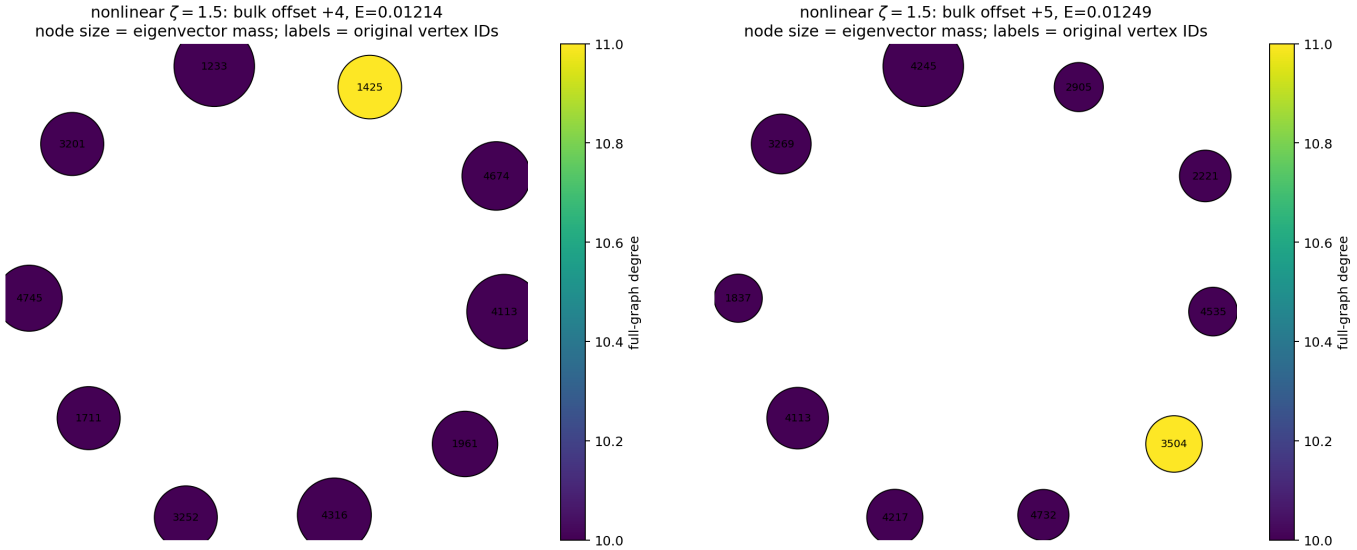


bulk offset p002



bulk offset p003

Figure 35: Original centered bulk induced supports.

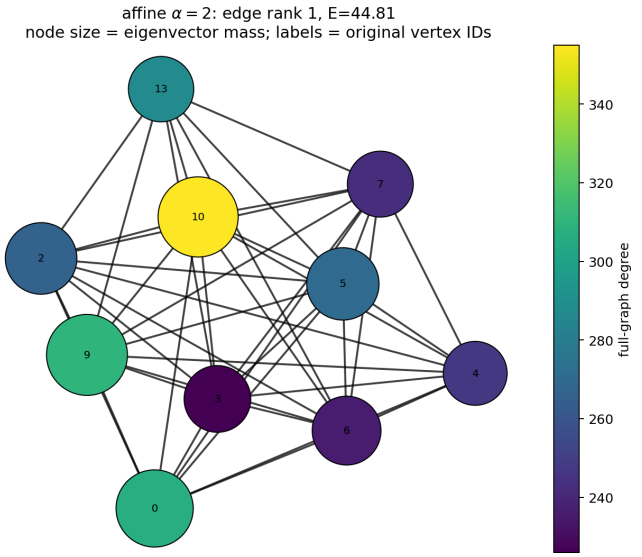


bulk offset p004

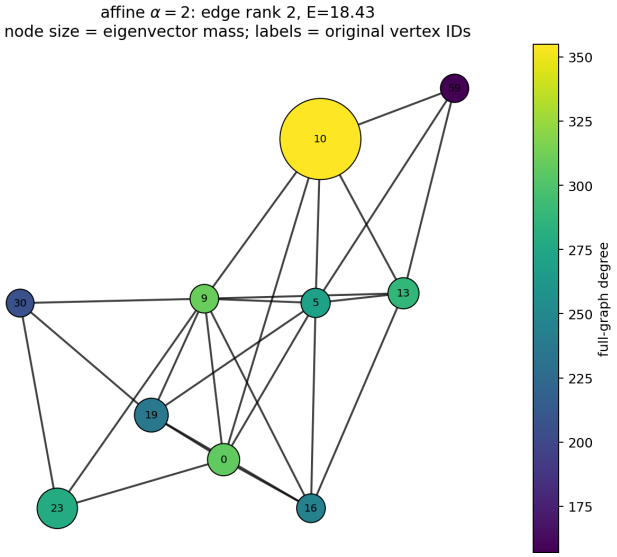
bulk offset p005

Figure 36: Original centered bulk induced supports.

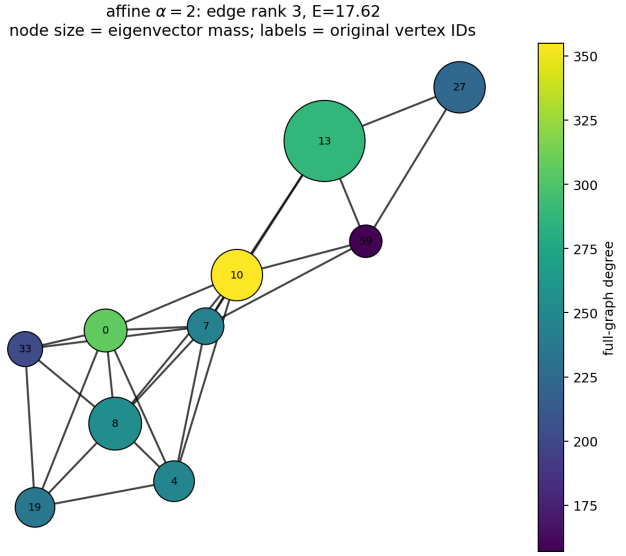
14.5 affine $\alpha = 2.0$: original edge ranks 1–10



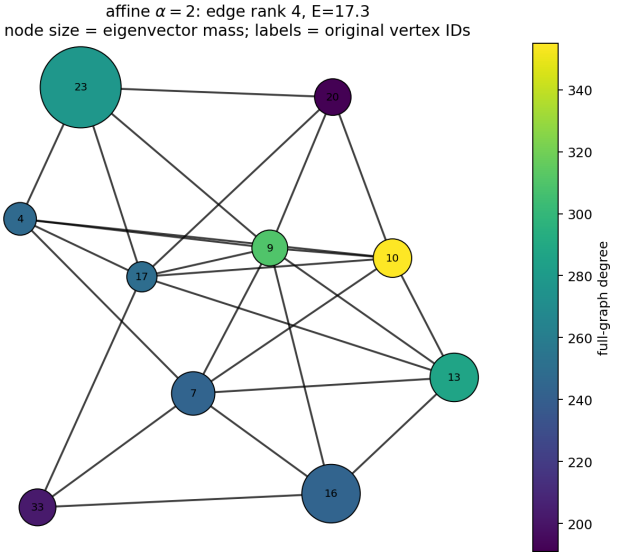
edge rank 001



edge rank 002

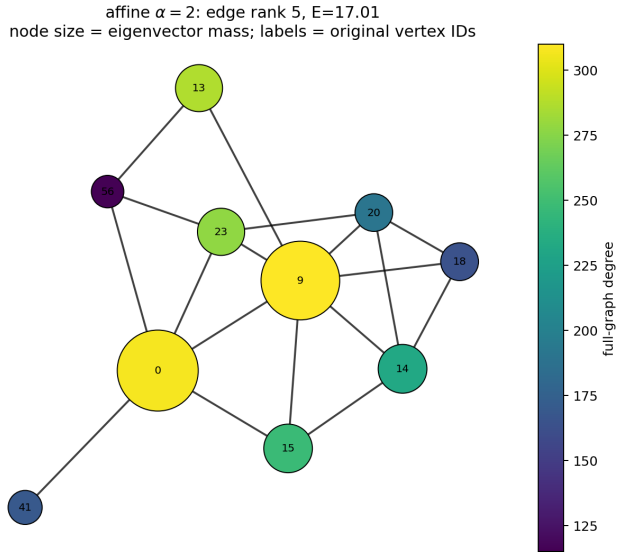


edge rank 003

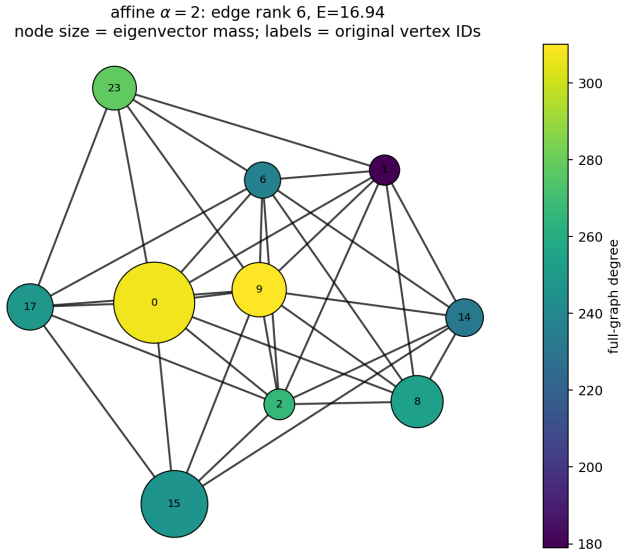


edge rank 004

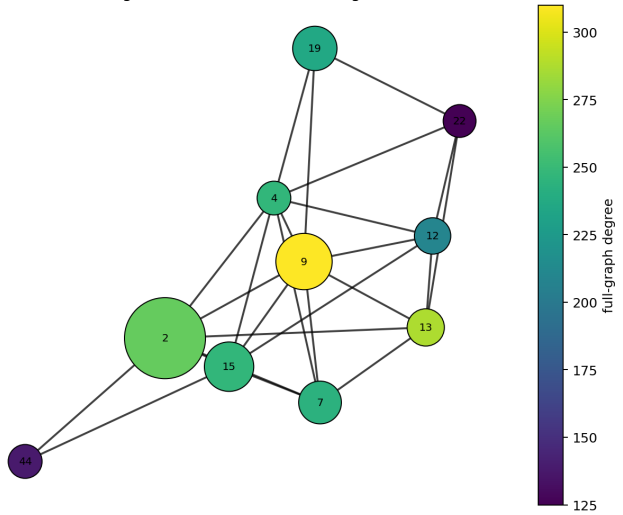
Figure 37: Original edge induced supports, ranks 1–4.



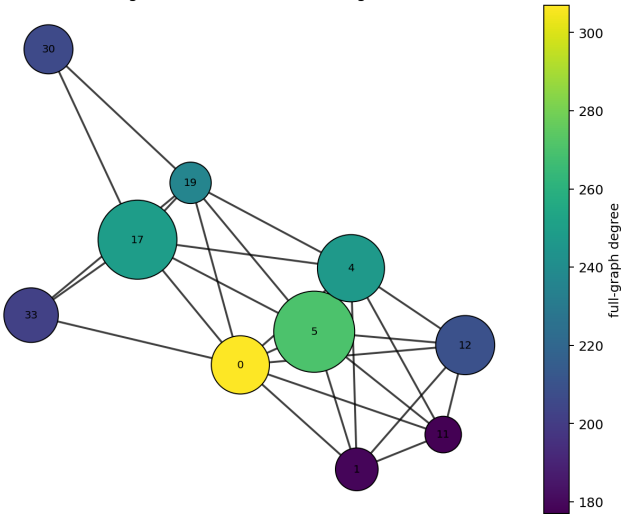
edge rank 005



edge rank 006

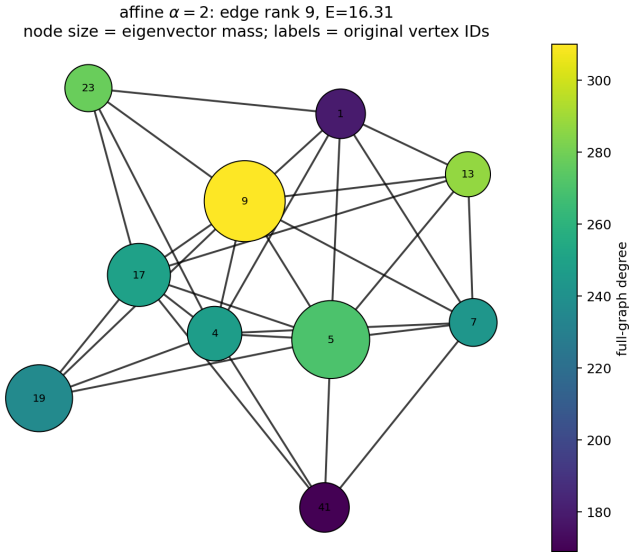


edge rank 007

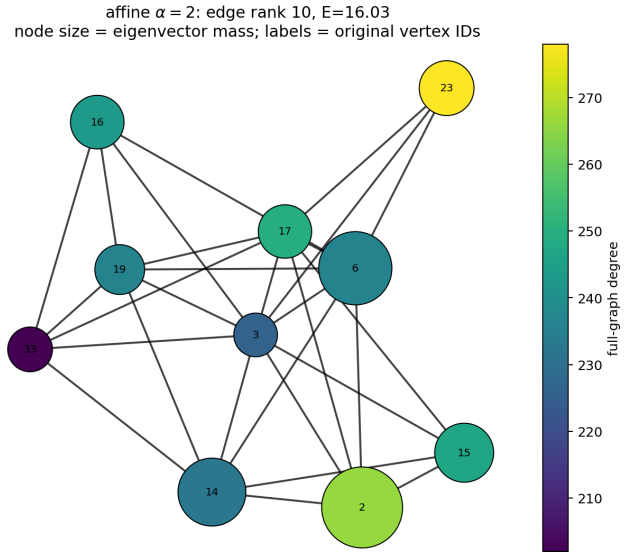


edge rank 008

Figure 38: Original edge induced supports, ranks 5–8.



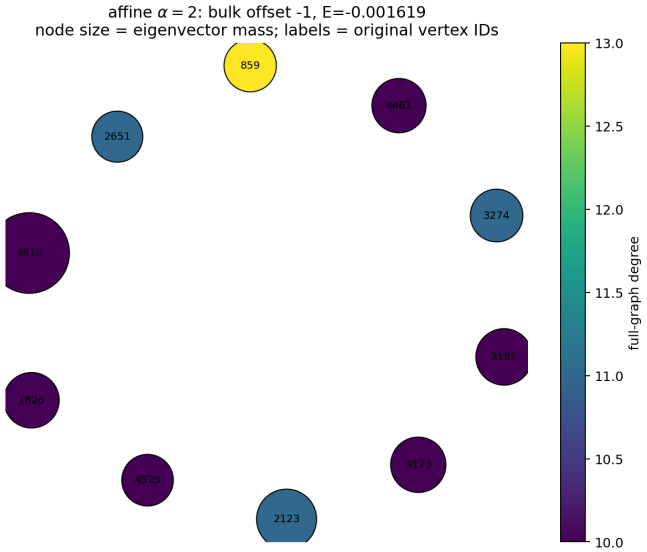
edge rank 009



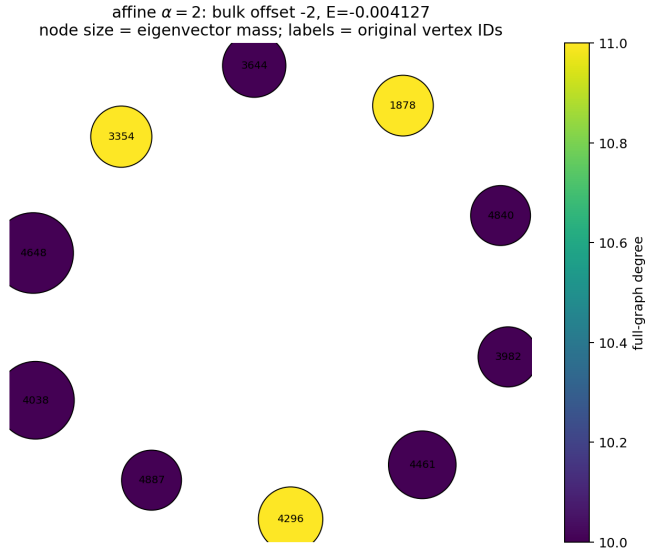
edge rank 010

Figure 39: Original edge induced supports, ranks 9–10.

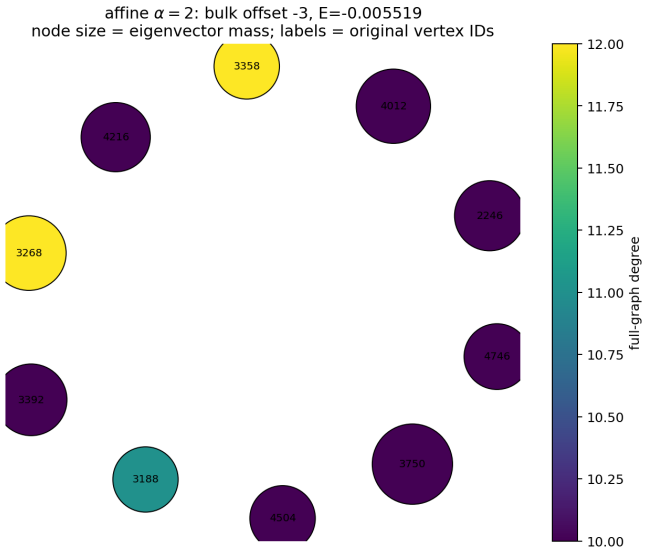
14.6 affine $\alpha = 2.0$: original centered bulk window



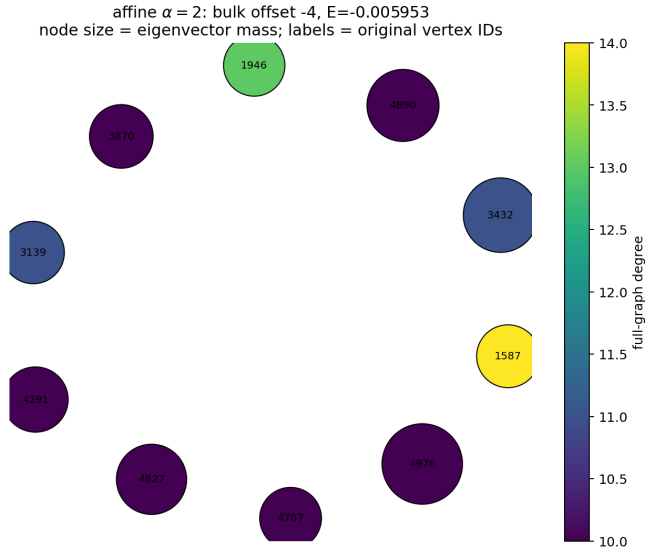
bulk offset m001



bulk offset m002

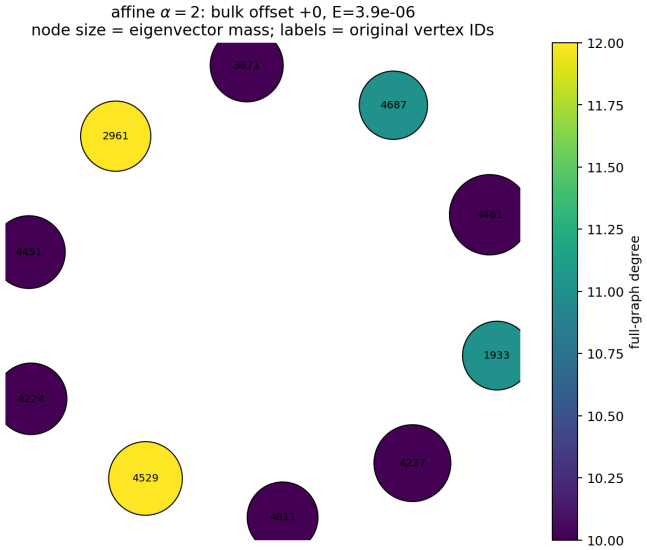


bulk offset m003

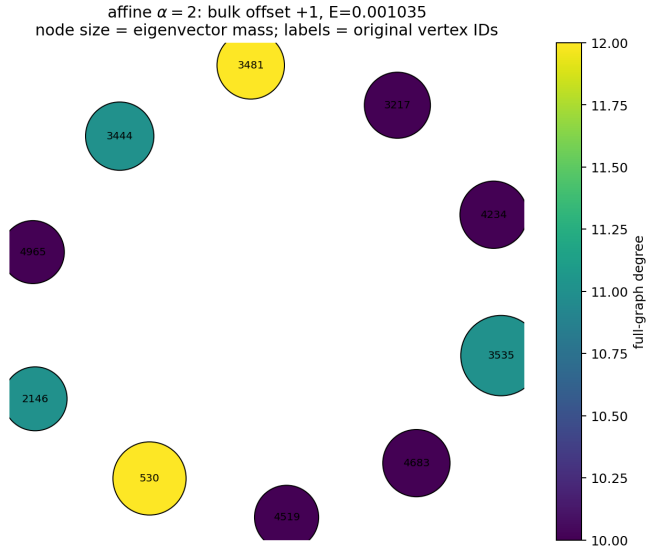


bulk offset m004

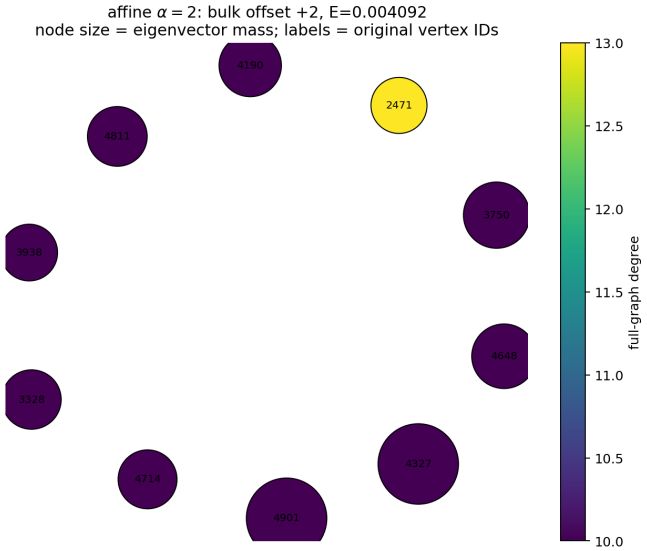
Figure 40: Original centered bulk induced supports.



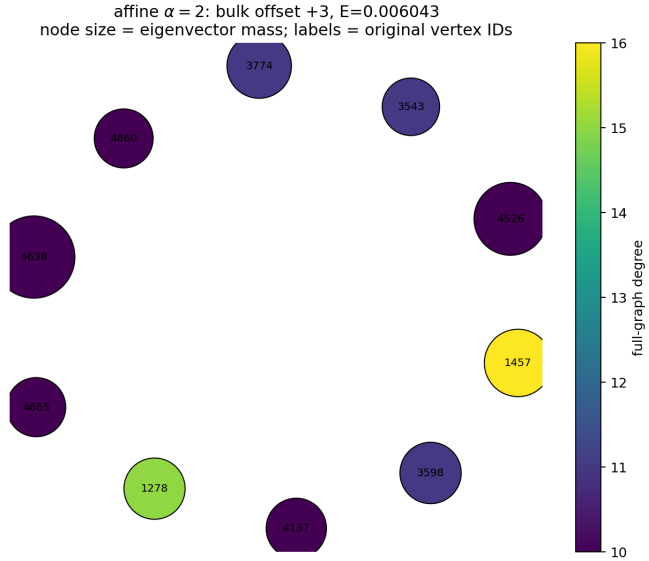
bulk offset p000



bulk offset p001

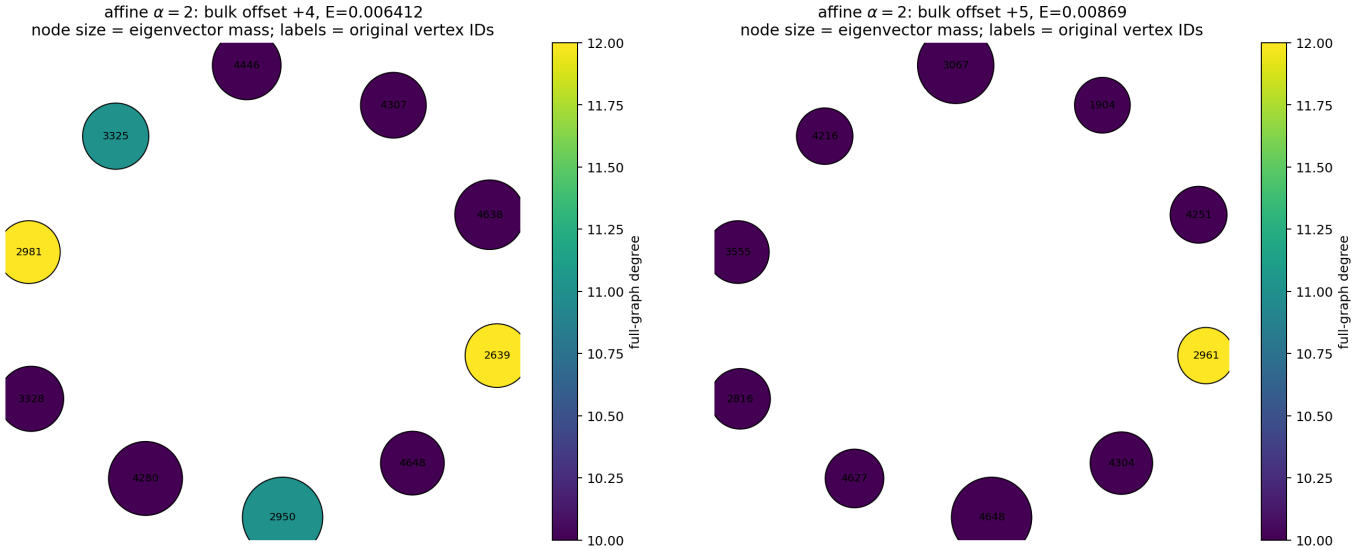


bulk offset p002



bulk offset p003

Figure 41: Original centered bulk induced supports.



bulk offset p004 bulk offset p005

Figure 42: Original centered bulk induced supports.

15 Explicit figure-file manifest

Every path below is quoted literally by an `\includegraphics` command and included in the package.

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figures/affine_alpha_2p0/histograms/single_graph_local_raw_spacing_histograms.png	327139
figures/affine_alpha_2p0/histograms/single_graph_local_unfolded_spacing_histograms.png	327407
figures/affine_alpha_2p0/induced_subgraphs/bulk/bulk_offset_m001.png	96117
figures/affine_alpha_2p0/induced_subgraphs/bulk/bulk_offset_m002.png	99902
figures/affine_alpha_2p0/induced_subgraphs/bulk/bulk_offset_m003.png	107905
figures/affine_alpha_2p0/induced_subgraphs/bulk/bulk_offset_m004.png	104920
figures/affine_alpha_2p0/induced_subgraphs/bulk/bulk_offset_p000.png	108098
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figures/affine_alpha_2p0/induced_subgraphs/bulk/bulk_offset_p002.png	97954
figures/affine_alpha_2p0/induced_subgraphs/bulk/bulk_offset_p003.png	96720
figures/affine_alpha_2p0/induced_subgraphs/bulk/bulk_offset_p004.png	107718
figures/affine_alpha_2p0/induced_subgraphs/bulk/bulk_offset_p005.png	102979
figures/affine_alpha_2p0/induced_subgraphs/edge/edge_rank_001.png	246894
figures/affine_alpha_2p0/induced_subgraphs/edge/edge_rank_002.png	156229
figures/affine_alpha_2p0/induced_subgraphs/edge/edge_rank_003.png	134637

TeX-relative filename	Bytes
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figures/affine_alpha_2p0/induced_subgraphs/edge/edge_rank_006.png	212213
figures/affine_alpha_2p0/induced_subgraphs/edge/edge_rank_007.png	142883
figures/affine_alpha_2p0/induced_subgraphs/edge/edge_rank_008.png	160499
figures/affine_alpha_2p0/induced_subgraphs/edge/edge_rank_009.png	189595
figures/affine_alpha_2p0/induced_subgraphs/edge/edge_rank_010.png	192340
figures/affine_alpha_2p0/spectrum/single_graph_eigenvalue_density.png	63285
figures/nonlinear_zeta_1p5/bulk/bulk_offset_diagnostics.png	256403
figures/nonlinear_zeta_1p5/edge/edge_rank_diagnostics.png	259680
figures/nonlinear_zeta_1p5/edge_vs_bulk/edge_vs_bulk_comparison.png	185800
figures/nonlinear_zeta_1p5/energy_ipr/single_graph_localization_vs_energy.png	142161
figures/nonlinear_zeta_1p5/histograms/single_graph_global_spacing_and_ratio.png	133472
figures/nonlinear_zeta_1p5/histograms/single_graph_local_raw_spacing_histograms.png	264631
figures/nonlinear_zeta_1p5/histograms/single_graph_local_unfolded_spacing_histograms.png	331044
figures/nonlinear_zeta_1p5/induced_subgraphs/bulk/bulk_offset_m001.png	109974
figures/nonlinear_zeta_1p5/induced_subgraphs/bulk/bulk_offset_m002.png	97730
figures/nonlinear_zeta_1p5/induced_subgraphs/bulk/bulk_offset_m003.png	110125
figures/nonlinear_zeta_1p5/induced_subgraphs/bulk/bulk_offset_m004.png	98946
figures/nonlinear_zeta_1p5/induced_subgraphs/bulk/bulk_offset_p000.png	99044
figures/nonlinear_zeta_1p5/induced_subgraphs/bulk/bulk_offset_p001.png	96876
figures/nonlinear_zeta_1p5/induced_subgraphs/bulk/bulk_offset_p002.png	94919
figures/nonlinear_zeta_1p5/induced_subgraphs/bulk/bulk_offset_p003.png	105800
figures/nonlinear_zeta_1p5/induced_subgraphs/bulk/bulk_offset_p004.png	96375
figures/nonlinear_zeta_1p5/induced_subgraphs/bulk/bulk_offset_p005.png	95024
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figures/nonlinear_zeta_1p5/induced_subgraphs/edge/edge_rank_003.png	293516
figures/nonlinear_zeta_1p5/induced_subgraphs/edge/edge_rank_004.png	185596
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figures/nonlinear_zeta_1p5/induced_subgraphs/edge/edge_rank_010.png	271762
figures/nonlinear_zeta_1p5/spectrum/single_graph_eigenvalue_density.png	78462

16 Machine-readable outputs and limitations

- `tables/window_summary.csv`: all means and medians.
- `tables/random_1000_summary.csv`: 1,000-set reference distributions.
- `tables/rank51_sparse_validation.csv`: disclosed extension validation.
- `figure_manifest.csv` and `FIGURES.md`: exact filenames and hashes.

No new graph ensemble, uncertainty analysis, finite-size scaling, degree-matched null, or parameter sweep was run.